Text properties and children's reading pace in a multilingual storytelling app

Abstract

Children increasingly read machine-generated stories on screens, and whether the text properties governing laboratory reading also govern it there is unknown. We will analyse logs from *Let's Story*, in the *Applaydu* app, where children aged 7 and over read in ten or more languages. Reading time per word will be modelled from word difficulty, grammatical load and how explicitly ideas are linked. Six confirmatory hypotheses concern these associations, their change with age, their selective weakening under narration, children's difficulty ratings and accumulated in-app reading. Each association, estimated with readings grouped by text, child and language, is judged against a pre-set smallest effect of interest. Associations clearly beyond it would support the hypotheses, in either direction for the two that separate rival developmental accounts, and those clearly below it would show that laboratory effects do not transfer. Because families choose narration, texts and continued use, every estimate is a conditional association.

Introduction

A generation of children now does part of its reading on a screen, and some of what appears there was written by a machine. In the United Kingdom, one in five 8- to 18-year-olds read fiction on screen at least monthly in 2025, and the share reading print fiction had fallen from three in five to one in two since 2017 (Picton & Clark, 2026). The same report found that children reading fiction only on screen were disproportionately those who enjoyed reading least. Commercial storytelling applications generate a narrative to order, read it aloud in the family's language and record what the child did with it. One such feature reported 2.4 million story sessions in its first six months (Ratner et al., 2025). Yet many parents interviewed in one small study wanted the plot and dialogue of a child's story to remain human-authored, accepting machine-made illustrations only on that condition (Jin & Yuan, 2025). The behavioural record such platforms accumulate far exceeds any laboratory corpus of children's reading, spans ten or more languages and arises under the conditions that laboratory work excludes. We ask whether the text properties that govern reading under controlled conditions still govern it outside the laboratory, and whether their hold on the reader changes over the years in which decoding becomes automatic. Either answer gives an estimate that no laboratory can supply. It concerns how far three determinants of reading established under experimental control extend to unsupervised reading of generated stories in ten or more languages, with the developmental change in each.

The laboratory findings are well established at three levels of linguistic description. The reading systems framework separates word identification, word-to-text integration and the situation model (Perfetti & Stafura, 2014). This study's three text components, LexicalDifficulty, GrammaticalLoad and ReferentialContinuity, are built to instantiate that partition, each derived from the text features at one level (word, sentence or passage).

At the level of the word, high-frequency words are recognised faster and fixated for less time than low-frequency words, and frequency and word-length effects on text reading are closely similar across 12 alphabetic languages (Kuperman et al., 2024). Word-length effects take a script-specific form outside the Roman alphabet (Lahoud et al., 2023), a caution that concerns only Korean in this sample. Children's eye movements are driven by the frequency of the words they meet (H. S. S. L. Joseph et al., 2013). Compared with adult controls, children showed larger length and frequency effects (Tiffin-Richards & Schroeder, 2015), as predicted by an account on which the frequency effect tracks accumulated exposure (Brysbaert et al., 2018). The same developmental pattern appears outside alphabetic scripts (Lian et al., 2026), and the Lexical Quality Hypothesis describes it (Perfetti, 2007; Perfetti & Stafura, 2014). A word is retrieved quickly and reliably in proportion to how well specified and tightly bound its orthographic, phonological and semantic constituents are, and that quality is built by reading itself (Castles et al., 2018; Nation, 2017). On that account, a text's lexical properties, a child's age band (7 to 8, or 9 and over) and accumulated reading all bear on the cost of identifying a word. Hypothesis 1 tests the account through text properties, and Hypothesis 6 through accumulated in-app reading.

A second source of word-level cost is contextual. On surprisal theory, a word costs time in proportion to how unexpected it is given what precedes it (Hale, 2001; Levy, 2008). The relation is close to linear in surprisal (Shain et al., 2024; Smith & Levy, 2013) and holds in 11 languages from five families (Wilcox et al., 2023). That linearity makes a text's mean per-word surprisal the natural

text-level summary of the cost, and the design adopts it, allowing for two qualifications. A text's overall difficulty moderates how strongly per-word surprisal bears on reading time, so the text-level mean is not a neutral summary of a constant word-level effect (Chen et al., 2026). Frequency and surprisal are hard to separate, since dissociable frequency effects shrink as surprisal predictors improve (Oh & Schuler, 2025), which qualifies their reported additivity in naturalistic reading (Shain, 2024). LexicalDifficulty pools the two, so the confirmatory claim concerns their joint contribution and leaves the share of each unresolved. How subword tokens are aggregated to words affects psycholinguistic estimates (Giulianelli et al., 2024), and the unit over which surprisal should be defined is being reconsidered (Kiegeland et al., 2026), so the aggregation rule is fixed under Materials. In children, contextual facilitation appears in the later eye-movement measures that a whole-text duration absorbs (Tiffin-Richards & Schroeder, 2020).

Further lexical properties enter the design, each for a different reason. Concrete words are fixated for less time than abstract ones in natural reading, on every fixation measure where length and frequency were matched (Yao et al., 2024) and on first-fixation duration where they were covaried instead (Magnabosco & Hauk, 2024). That advantage is usually traced to perceptual grounding, on accounts that treat conceptual processing as partly modal and partly amodal (Kaup et al., 2025). A word's age of acquisition bears on how quickly it is identified over and above its frequency (Juhasz, 2005). Function words are skipped far more often than content words (Rayner, 1998), so a text of higher lexical density (the proportion of content-word tokens) presents more tokens to be fixated and identified. Reading times for a word fall over repeated encounters, in children of 10 and 11 reading sentences (H. Joseph & Nation, 2018), so a lexical diversity index (McCarthy & Jarvis, 2010) tracks the share of tokens a reader meets for the first time. A word-level measure of informational cost predicts eye movements in first- and second-language readers of diverse writing systems (Kimchi et al., 2025), though its authors test no text-level aggregate of the kind used here.

At the level of the sentence, structure imposes a further cost. Mean dependency distance indexes the distance over which a dependency must be resolved, and clause density, subordination and noun-phrase depth index how many dependencies are open at once. Structural cost is not reducible to predictability. In 368 adults reading syntactically challenging sentences, language-model surprisal from 409 estimators approximated the early effects of disambiguation but not the additional rereading that repairing a structure incurred (Timkey et al., 2026). For that reason, the grammatical features form a distinct family and are never absorbed into surprisal. Gibson's (1998) theory separates the cost of keeping a prediction alive from that of integrating the element that completes it. Retrieval-based accounts place the cost at the point of resolution, where intervening and similar material slows retrieval (Lewis & Vasishth, 2005). Expectation-based accounts add that the material before a head can make the head more predictable (Levy, 2008), so in head-final configurations a longer dependency need not cost more. A clause-final verb has been read faster after longer intervening material in German and Hindi (Konieczny, 2000; Vasishth & Lewis, 2006), and the strength of a locality effect is expected to vary with a language's statistics (Futrell et al., 2020). The grammatical slope is therefore allowed to vary by language in every model below.

Hypothesis 2 asks whether children bear these costs disproportionately. A prior question is whether a grammatical cost appears at all when reading time is measured over whole texts, as here. In naturalistic corpus reading by adults, a distance-based integration cost predicted reading times at nouns and auxiliaries but not at words in general, whereas word predictability predicted them

throughout (Demberg & Keller, 2008). In the Multilingual Eye-movement Corpus, global measures of syntactic and lexical complexity (among them the longest dependency path, the proportion of embedded sentences and mean sentence length) predicted adults' passage reading negligibly once passage length was accounted for (Kuperman et al., 2025). At the sentence level, that study found effects of any size only for sentence length and position. Its lexical measures proved as negligible as its syntactic ones, but its authors separate such global measures from the aggregate properties of a passage's individual words. LexicalDifficulty is anchored on those aggregates (mean frequency, word length and surprisal), whose word-level effects are well replicated across languages (Kuperman et al., 2024; Wilcox et al., 2023). The grammatical question remains open for children reading whole stories, and Hypothesis 1 is framed so that either answer is informative.

At the level of the passage, the property of interest is whether the text states the connection between one idea and the next or leaves the reader to supply it. Sentence-to-sentence semantic similarity, overlap in referring expressions and the density of explicit connectives together index how much of that work the text's surface has already done (Crossley et al., 2019). These are measures of cohesion, a property of the text, whereas coherence is a property of the representation the reader builds, and cohesion eases its construction without guaranteeing it (McNamara et al., 2010). Referential overlap descends from the rule that connects propositions sharing an argument (Kintsch & Dijk, 1978), which the construction-integration model retains in the weights it assigns to propositions close together in the text base (Kintsch, 1988). Semantic similarity between adjoining sentences approximates that propositional analysis (Foltz et al., 1998). An explicit relational marker speeds processing of the segment it introduces without changing what is recalled (Sanders & Noordman, 2000). Multi-dimensional characterisations of difficulty built from such indices outperform classical readability formulae (Crossley et al., 2023). Evaluated against eye movements, though, readability scorers account for real-time reading ease only partially (Gruteke Klein et al., 2026). Partly for that reason, this study models the psycholinguistic features and uses no composite readability score.

The systematic review accompanying this protocol refines these laboratory findings in ways that shape the design, and Supplementary Section S2 tabulates the studies closest to this study in method and modality. Text difficulty does not reduce to word frequency. In German fourth graders, estimated lexical coverage predicted comprehension non-linearly (Ludewig et al., 2023). In 158 children aged 8 to 11, word frequency, pronoun density, concreteness and deep cohesion each explained unique variance in passage comprehension once decoding and oral language were controlled, and lexical arousal explained variance beyond them (Pickren et al., 2022). That result warrants including affective and discourse variables alongside conventional readability measures. Syntactic complexity impaired comprehension accuracy and slowed recall in bilingual children (Wasito Kirana et al., 2025). In 5,133 second graders reading informational text, word concreteness, referential cohesion and deep cohesion each predicted oral reading rate beyond classical readability formulae (Tortorelli, 2020). In first graders, a text-level vocabulary aggregate predicted rate, accuracy and comprehension (Bruner et al., 2026). Both samples are younger than this study's age bands and read aloud, so they establish an association between text properties and reading rate without fixing its size for silent reading at 7 and over.

Two further strands concern the reader and the reading context. On the reader's side, longitudinal work shows that a child's reading history is not a simple quantity of practice. In random-intercept cross-lagged models from ages 7 to 16, weaker early reading predicted less subsequent leisure

reading, while later within-person increases in book reading predicted gains in comprehension (Torppa et al., 2020). A 10-year study likewise found reciprocal associations between skill and print exposure, with the skill-to-practice path most visible early and accumulated exposure more predictive later (van Bergen et al., 2021). Accumulated use of an app reflects both practice and the stable differences in skill and motivation that produced the use, so Hypothesis 6 separates within-child change from between-child differences. The setting matters as well as the quantity. Digital leisure reading and print reading do not relate to comprehension in the same way (Altamura et al., 2025), and digital leisure reading is associated with comprehension independently of print (Altamura et al., 2026). Hypothesis 6 counts only reading inside one app.

As for the reading context, read-aloud narration is often treated as a uniform benefit, but the evidence does not support that view. A meta-analysis of 30 reading-while-listening studies, 62 effects over a total N of 1,945, found a small pooled comprehension advantage ($g = 0.18$). There was no reliable advantage where ordinary reading was self-paced ($g = 0.06$, 95 per cent CI [-0.07, 0.19]), and a larger one only where the experimenter paced it ($g = 0.41$, 95 per cent CI [0.13, 0.70]) (Clinton-Lisell, 2023). Among students with reading difficulties, text-to-speech tools showed a moderate pooled effect ($d = 0.35$) that fell to $d = 0.15$, with an interval including zero, in within-participant designs (Wood et al., 2018). Neither synthesis identifies the mechanism, which a large naturalistic corpus is well placed to address. A spoken rendition may work by supplying word identity, in which case it should remove much of the cost of lexical difficulty while leaving the cost of connecting ideas intact. Alternatively, the audio's duration may set narrated reading time. Hypothesis 4 is designed to distinguish these decoding-scaffolding and audio-pacing accounts.

The generated stories studied here are a population of stimuli produced by one process, and a process need not distribute text properties as human authorship does. Comparisons in other domains report that large-language-model output has a distinct noun-heavy, informationally dense register even when prompted otherwise (Reinhart et al., 2025). Human writing in those comparisons shows more scattered sentence-length distributions and more varied vocabulary than model output (Muñoz-Ortiz et al., 2024). In creative writing, the domain closest to these stories, stylometric comparison separates the two on measurable features of style (O'Sullivan, 2025). Whether the lexical, grammatical and discourse determinants established on human-written text describe reading across such a population is a question about the scope of those findings (Haeffel & Cobb, 2022; Yarkoni, 2022).

To gauge how much published work combines this study's topics, we searched Scopus on 26 August 2026 for records from 2018 onwards. The search returned 3,701 records on children's reading comprehension, 6,592 on computational text features and 133,132 on platform log data. The pairwise intersections were 17, 28 and 45 records, and the three-way intersection was empty. The search covered one database under one set of expressions and missed the log analysis of this platform described below, although it had appeared two months earlier. The empty cell shows only that the intersection is thin and unevenly indexed, not that it is unoccupied. Figure 1 shows the sets, and Supplementary Section S4 records the search expressions and the publication trend. The search script, which archives the counts, is deposited with this protocol (Bernabeu, 2026c).

Over the same window, records combining children's reading with machine-generated text went from none before 2024 to two in 2024, six in 2025 and four in the first eight months of 2026. The studies are classroom experiments that report gains in comprehension and motivation but describe the texts only qualitatively (Balantekin, 2026; Bulut & Aktepe, 2026). Generated and human-written

texts have been compared directly, though not yet in children reading for themselves. Generated stories had fewer personal pronouns and more positive emotion than 100 human-written ones, and adult readers were less transported by them (Appel et al., 2025). The models tested for children's stories could not hold their vocabulary to a level appropriate for younger readers (M. Valentini et al., 2023), a limitation that bears directly on the lexical component.

Let's Story, a feature of the *Applaydu* app developed by Gameloft and Ferrero International, combines the three topics of the Scopus search. Families choose a setting, a hero, a companion, a plot and a theme for each generated story, and a generative model writes a narrative around those choices (Ratner et al., 2025). The app presents the narrative as an illustrated storybook, reads it aloud at the family's option (Ratner et al., 2025) and logs every completed reading. Comparable platforms have been studied one feature at a time. Booton et al. (2023) reviewed 11 studies and found inbuilt narration to aid comprehension and word learning. Log data from a research-oriented platform track ability, with reading speed in the BookBites app correlating with standardised scores at $r = .39$ to .48 (Sortkær et al., 2021). The commercial edutainment sector, with a far larger audience, is described in the literature we identified mainly by content analyses and small experiments (Chatzopoulos et al., 2023; Furlong et al., 2025).

One analysis of logs from *Let's Story* itself has been published. Over 419,700 story sessions in 18 languages across the 30 days following the feature's launch in November 2024, Ratner & Murphy (2026) describe how children aged 4 to 9 meet challenge. They report completion, the post-reading difficulty rating and where in a story abandoned sessions stopped. The analysis is descriptive by design and drawn from aggregated dashboard counts, so its authors state that it cannot follow an individual reader. Because it models no reading time and its summaries lacked the session-level structure a within-child term requires, it leaves every association registered here untested. Its window was the first month after launch, whereas ours opens more than a year later, so the novelty effects its authors mention apply much less here. The study does establish that the difficulty rating, the completion requirement and the session record can be analysed in large samples on this platform.

The theoretical commitments behind the hypotheses are set out in a machine-checkable theory object deposited with this protocol. It names twelve constructs, eighteen propositions, six auxiliary assumptions and six rival accounts. Each proposition states its mechanism, and each of the ten predictions is recorded with the proposition it derives from and, for seven, the rival it is diagnostic against. The object claims that the time a child spends per word is set by the costs a text imposes at the word, the sentence and the passage. Three reader capacities reduce those costs: the automaticity of print-to-sound mapping, verbal working memory and the readiness to supply an unstated connection. A spoken rendition bears more on the first than on the third. The platform records none of the three, and the age band stands in for them. Five further propositions extend the same constructs to the quality of the mental model a child ends with. They could be examined only in exploratory analyses at Stage 2, and only if comprehension items prove to exist, as set out under Contingent hypotheses. Figure 2, generated from the object, draws its constructs and the paths between them. The object satisfies all twelve items of the versioned rigour checklist in theoryforge, scoring 90.7 out of 100 (Supplementary Section S1) (Bernabeu, 2026d). Its causal graph is acyclic, so the independence claims it entails can be derived formally. Before interpreting the confirmatory models, we will test three of them, listed with their criterion under Outcome-neutral quality checks, and report any failure.

This study is observational throughout. Families choose the stories, the reading level and whether narration is on, and decide whether to keep using the app. A laboratory experiment establishes that a text property can change reading when an experimenter controls the text, the setting and the task. The further question, whether the property does so when nothing is controlled, concerns generalisation and cannot be answered in the setting that produced the original result (Nastase et al., 2020; Yarkoni, 2022). Whether a relation holds in a new population of texts, languages and readers therefore tests the account that predicts it (Haeffel & Cobb, 2022). In the logs of a vocabulary-learning app, a laboratory effect of valence held, while an effect of concreteness did not (Wild & Kuperman, 2024), though those readers were adults and the outcome was learning.

The platform offers three gains in return for that loss of control, beginning with stimulus sampling. Reading research has known since Clark (1973) that treating a handful of experimenter-written texts as a fixed set inflates confidence in conclusions meant to hold for texts in general. The mixed-model solution has been available for over a decade (Judd et al., 2012). The design treats Text as a crossed random factor over hundreds of texts that families selected, which is both the prescribed remedy and, in Brunswik's sense, a representative design (Dhami et al., 2004). A varying intercept does not by itself make an inference general where the variation between texts is systematic (Bear & Phillips, 2022), so the model includes the systematic sources of variation the platform records. Because an average over texts can hide the variation between them (Simonsohn et al., 2025), the text-level deviations will be displayed against the component scores at Stage 2. We avoid the phrase *ecological validity*, used in at least two senses, Brunswik's and Orne's (Kihlstrom, 2021), and argued to be too ill-defined to bear on generalisation at all (Holleman et al., 2020). Our claim is the narrower, checkable one about how the stimuli were sampled.

The second gain, linguistic breadth, matters because English dominates cognitive science to a degree that warps which mechanisms are studied and licenses overgeneralisation from one atypical language to the species (Blasi et al., 2022). Reading is among the worst affected, because writing systems differ in ways that matter for every stage of processing. The most ambitious response to date, the Multilingual Eye-movement Corpus, assembled eye-tracking during text reading in 13 languages, since extended to 21 (Siegelman et al., 2022; Siegelman et al., 2025). Its authors report that the existing empirical base is insufficient for cross-linguistic comparison. A commercial platform covers many of the same languages incidentally, with far more readers per language, and its readers are children, whereas the corpus records skilled adults. The platform records only whole-story durations, so it gains breadth at the cost of the resolution an eye-movement record gives.

The direction of each of the three associations is expected in all ten languages (Dutch, English, French, German, Greek, Italian, Korean, Polish, Portuguese and Spanish). The operations reading recruits, mapping a written form to sound and to meaning and relating it to what came before, are common to writing systems that differ in almost everything else (Frost, 2012; Verhoeven & Perfetti, 2022). The adult evidence bears this out for word length and frequency in 12 alphabetic languages (Kuperman et al., 2024). In magnitude, by contrast, the lexical association is expected to vary with how a script encodes the word, and the grammatical association with a language's word order and statistics. No comparable reason has been derived for referential continuity, which operates above the word. These expectations concern the language-varying component slopes, which are reported for every language, though no verdict in Table 1 rests on any one of them.

The third gain is behaviour logged by the app as it occurred. Logged and self-reported digital media

use diverge systematically enough to have warranted a meta-analysis of the discrepancy (Parry et al., 2021). Among 109 French primary-school children, parents' estimates of weekly reading time were three times the logged figure, which predicted fluency better (Brossette et al., 2026).

Against these gains stands the strongest statistical objection to any large non-random dataset. Meng's (2018) decomposition shows that the error of a sample mean is the product of three quantities: data quality, data quantity and the difficulty of the problem. Data quality is the correlation between the recorded value and whether it was recorded at all, and probability sampling keeps it near zero. Once it is no longer near zero, the population size stops cancelling, and estimation error grows with the square root of the population for any given sample size. Bradley and colleagues (2021) gave the canonical demonstration. Two surveys collecting about 250,000 responses a week and 75,000 a fortnight overestimated United States COVID-19 vaccine uptake by 17 and 14 percentage points, with margins of error so small that the wrong answers looked precise. A well-designed panel of about 1,000 responses a week, by contrast, was accurate. Every point of selection here is a candidate for that correlation: who owns a tablet, who installs an app tied to a confectionery brand, who registers consent for research use, who switches narration off and who finishes a story.

Meng's result, however, concerns the estimation of a population mean, and this study estimates none. Its estimands are conditional associations inside the sample, within a language and within a reading level. A point of selection that depends on the reader alone, such as who owns a tablet or who registers consent, changes which children a slope describes without distorting it. Where the reader attribute also moderates the slope, the estimate becomes the average slope across the families who use the platform. An estimate of a heterogeneous association from a non-random sample should be interpreted as such an average (Bryan et al., 2021). We will report these coefficients as estimates for children who read on this platform, never as population values for children in general, and will report no population mean of reading speed for any group. A slope is distorted only when selection is a common effect of the text property, or of something correlated with it, and of the outcome or of an unmeasured cause of the outcome (Elwert & Winship, 2014; Hernán et al., 2004; Lamp & MacKinnon, 2025). That condition is narrower than the one that biases a mean. Of the points of selection listed above, only completion meets it. Narration is a state the models condition on, so the selection it involves affects the denominators of the Hypothesis 4 proportions and is addressed with that hypothesis.

The extract that the app's data controller will transfer contains completed readings only, so the analysis conditions on completion, which is plausibly a common effect of the text's difficulty and the reader's capacity. Conditioning on a common effect can induce an association between its causes where none existed (Rohrer, 2018), and the direction here is predictable. If harder texts are abandoned more often, the hard texts that remain are disproportionately those read by children who could manage them, which attenuates the coefficients Hypothesis 1 predicts. If younger children abandon demanding texts more often, their remaining readings of such texts are the more selected and their slopes the more attenuated (Elwert & Winship, 2014; Hernán & Monge, 2023). On the sign convention given below, that attenuation pushes the grammatical interaction positive and the continuity interaction negative, towards the bottleneck-release and cue-utilisation accounts of Hypotheses 2 and 3 respectively. The bias therefore works against our predictions, so a non-negligible result withstands the objection better than a null does. Where the data controller can supply the completion rate by age band, we will report it with every verdict on Hypotheses 1 to 3. If

the younger band's rate is the lower, a verdict supporting the bottleneck-release or cue-utilisation account will be reported as also consistent with differential attrition.

A reader can cope with a demanding text without abandoning it, by reading it less carefully. When reading speed was manipulated, comprehension held across a broad range of rates and fell only at the fastest, around 405 words per minute, while the influence of word properties on eye movements persisted throughout (Schwalm et al., 2026). That result places the 400-words-per-minute plausibility bound close to the rate at which comprehension gave way, so the bound removes page-flipping but keeps fast, still comprehending reading. The same study gives no warrant for assuming that faster readings show weaker text effects, since its word-level effects did not weaken with speed. Neither the extract nor the design can separate a skimmed reading from a fluent one, so we register the ambiguity without assigning it a direction.

We do not use causal verbs for variables that families chose. The questions are nonetheless causal, and we say so because euphemism has been argued to damage observational research by obscuring what is being asked (Grosz et al., 2020; Hernán, 2018). We state the causal structure in the theory object's causal graph, derive the adjustment set from it, register the rival explanations it implies and report associations conditional on that adjustment.

Secondary data invite the analytic flexibility that preregistration exists to remove, and the field has standards for preregistering analyses of existing data (Van den Akker et al., 2021; Weston et al., 2019). The arrangement described under Data provenance goes further, since the data controller holds the extract, unanalysed, and will transfer it only after in-principle acceptance. As a result, the most common objection to observational work, that the analyst saw the data before choosing the analysis, does not arise. Withholding the outcome until the analysis is fixed is the principle of blind analysis (Dutilh et al., 2021; MacCoun & Perlmutter, 2015).

The analysis estimates a few pre-specified associations and does not attempt to predict an individual child's reading. We calculated the precision attainable for each without access to the extract and state in advance which of them the available records cannot resolve. Group-level associations need not describe any individual, since variance around an expected value was two to four times larger within people than within groups (Fisher et al., 2018). The within-child analysis uses tier 2, the user accounts with at least three eligible readings of distinct texts, to separate each child's own change from differences between children. The two are never reported as a single quantity.

The design allows for three features of the platform and its records. First, the reading level a family sets determines which version of a text the platform presents, so text difficulty is partly assigned by a choice that also reflects the child's capacity. We will estimate the text coefficients within reading level and check whether they persist under that restriction. Second, the outcome is a whole-story rate, and we state below what such a rate can and cannot index. Third, the data contain no measure of a child's reading skill, vocabulary or language background. The reading language a family selects need not be the child's first language, so every coefficient below averages over children reading in a first and in a second language. In adult bilinguals reading paragraphs, frequency effects were larger in the second language than in the first and grew as second-language exposure fell (Whitford & Titone, 2012). A reading language is not a language background, and the country-by-language analysis registered under Sensitivity analyses is the extract's only means of addressing the difference. Without measures of skill and vocabulary, the age band stands in for a

bundle of developing capacities.

Accounts of reading development suggest what the age band can index when no measure of skill is available. Under the simple view, reading comprehension is the product of decoding and language comprehension, and decoding's share of the variance falls across the primary grades (Gough & Tunmer, 1986; Hoover & Gough, 1990; Lonigan et al., 2018). On automaticity theory, that share falls because identification that runs without deliberate attention releases capacity for integration (LaBerge & Samuels, 1974; Perfetti, 2007). Phonological recoding on each new word is the mechanism by which reading itself produces that automaticity (Share, 1995), which ties Hypothesis 6 to that theory. In a longitudinal eye-movement study, reading efficiency rose by 36 per cent between grades 2 and 4, and linguistic skill determined the change whereas oculomotor skill did not (Huestegge et al., 2009). Among 194 Greek-speaking children, silent reading speed rose from 111 words per minute in grade 3 to 166 in grade 6, through fewer and shorter fixations and longer saccades (Gleni et al., 2026). In the same study, those oculomotor parameters explained variance in speed beyond vocabulary and cognitive measures. The age band is therefore interpreted as a coarse, aggregate proxy for reading experience and the automaticity it confers, not for any single capacity.

The platform sets the boundary between the age bands. It describes its audience as children of 4 to 9 (Ratner & Murphy, 2026), so the older band is expected to lie mostly at its lower edge, and its upper edge is not recorded. Whether change across such a boundary is governed by attained reading level or maturing oculomotor control is still debated (Justino & Kolinsky, 2023; Louwen et al., 2026). These data can address that question only through the finer-age sensitivity analysis registered below.

Every hypothesis but Hypothesis 5 has the same outcome, the time a child spends per word over a whole story. In the laboratory, reading time is informative about processing because a word is fixated for about as long as it takes to process (Just & Carpenter, 1980). The properties that lengthen fixations predict passage-level eye-movement measures on texts of more than 200 words, a relation shown so far in second-language readers of English (Nahatame & Yamaguchi, 2025). The theory object records this inference as the auxiliary assumption that elapsed time indexes effort, on which six of its ten predictions depend. The positive control on recorded reading time, registered under Outcome-neutral quality checks, puts that assumption at risk. A whole-text rate is more than the sum of its word-level costs, since the efficiency of processing items in a series explained variance in reading connected text once discrete word reading was controlled (Romero et al., 2024). The components are properties of the text, so differences in readers' efficiency are absorbed by the age band and the residual. We describe the outcome as reading pace, an index of the processing demand a text imposes, and not as a measure of comprehension or fluency, because nothing in these data supports either description.

The theory object registers one rival interpretation of the outcome, the engagement account. It traces time on a story to how engaging the story is, whereas this study traces it to how demanding the text is. Among 122 sixth graders reading silently, passage difficulty, rated situational interest and their interaction all predicted time on task, and interest accounted for the most variance (Welland & Burns, 2025). Two registered predictions are marked diagnostic against the engagement account. For the continuity coefficient, that account would produce a non-negligible positive value, since a text that states more of its connections might hold a child longer. The theory predicts instead that such a text should let the child finish sooner. Hypothesis 5 is also diagnostic, because a difficulty rating formed from effort the child noticed expending should rank the three components as reading time does, but

a rating tracking enjoyment need not. Neither prediction alone rules out the engagement account.

Two of the six confirmatory hypotheses below concern interactions between a text property and the age band. Both are registered as two-sided tests that adjudicate between rival accounts, because the direct developmental evidence in this age range supports neither sign.

The age band and narration enter the models as deviations from their observed prevalence in tier 1 (every eligible completed reading). A reading scores 1 for the older band, or for narration on, and 0 otherwise, minus the proportion of tier-1 readings in that state. Each component's main effect is then its prevalence-weighted average slope across the two states, free of design effects from imbalance. Each interaction is the difference in slope between the two states, positive when the slope is larger in the older band or under narration. The reading-time outcome is log-transformed and standardised within language, so a coefficient is a within-language standardised slope. The region of practical equivalence, the range of effects treated as negligible, has a half-width of 0.05 standard deviations for such a slope. Hypothesis 4 concerns a difference between two attenuation proportions, with a region of half-width 0.05 on that difference. Hypothesis 5 uses the child's own three-point difficulty rating, and its region has a half-width of log(1.05) on the log-odds scale, corresponding to odds ratios between 0.952 and 1.05.

LexicalDifficulty and GrammaticalLoad are oriented so that higher scores mean a more demanding text, and ReferentialContinuity so that higher scores mean a text that states more of the links between its ideas. The orientations are fixed before any component score is linked to outcome data. Because the main effects of grammatical load and referential continuity run in opposite directions, the same developmental claim takes opposite signs in the two interactions. Grammatical load is associated with slower reading, so a weaker association in older children makes a positive slope smaller and the interaction negative. Referential continuity is associated with faster reading, so a weaker association in older children makes a negative slope less negative and the interaction positive.

Children in the two age bands differ in overall reading rate, and a purely multiplicative difference in rate produces a spurious interaction on a raw time scale (Faust et al., 1999). On the log scale, such a difference appears only as a main effect, so every interaction reported here is a difference in *proportional* cost.

The log transformation leaves two other sources of interaction in place. First, an interaction can arise from an additive component of the interval. If the time a word costs is a base decoding time plus a grammatical increment, the log-scale grammatical slope is that increment divided by the base. A reader with a smaller base then shows a larger proportional slope even when the increment is identical. The bottleneck-release account posits an additive decoding component of this kind, so a positive Hypothesis 2 interaction is that account's expected signature on this scale, not an artefact of the transformation. Second, part of any whole-story interval is spent on the illustrations, on turning pages and away from the task. Children's attention to the text of an illustrated book rises across the primary grades (Roy-Charland et al., 2007), so the non-reading share may be larger in the younger band. The younger band's proportional slopes would then be smaller, moving the Hypothesis 2 interaction towards the positive sign and the Hypothesis 3 interaction towards the negative one. These data cannot show whether a smaller base interval reflects faster decoding or less time away from the text. A practically non-negligible interaction of either sign will be reported with that limitation.

The remaining age-band interaction, LexicalDifficulty × AgeBand, has no registered test and no signed expectation. The eye-movement literature establishes larger absolute lexical effects in children (Tiffin-Richards & Schroeder, 2015) without showing whether they are larger as a proportion of a slower baseline, the quantity a log outcome measures. The developmental evidence comes from raw latencies on isolated words. Across grades 2 to 4, word length and regularity effects declined while frequency effects persisted (Heß et al., 2026). In Italian children of grades 1 to 3, sensitivity to word length fell with age, in a study that manipulated length alone (Zoccolotti et al., 2005).

Expressing the outcome per word addresses one confound and leaves another. In the standard cohesion manipulation, the more cohesive version of a text is 20 to 40 per cent longer. A whole-text duration would then show higher cohesion slowing reading for arithmetic reasons alone. Dividing by word count and retaining log text length as a covariate closes that path, and a raw duration is never used as an outcome. A per-word rate still includes the extra time readers spend at sentence boundaries, time whose cause is uncertain (Rayner et al., 2000; Warren et al., 2009), and a text of shorter sentences has more boundaries per word. The boundary time works against the expected sign of sentence length taken alone.

Table 1 sets out each hypothesis against the sampling and analysis plan that tests it, the precision the design analysis calls for, how each possible outcome will be interpreted and what would be shown wrong. We fixed every row before we had any access to outcome data.

Table 1

Study design table. ROPE denotes the region of practical equivalence and SD *a standard deviation. A range in brackets after a requirement is its 95 per cent interval. The base eligibility rules, set out in the Methods under Sampling plan, apply to every row. No datum is ever replaced.*

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Are the text properties that govern reading in the laboratory associated with reading pace in an app?	H1. LexicalDifficulty and GrammaticalLoad are positively, and ReferentialContinuity negatively, associated with log reading time per word.	Tier 1: every eligible completed reading in the frozen extract, one per account and text, from accounts recorded in either age band, 7 to 8 or 9 and over. There is no stopping rule, because the extract is fixed before modelling. ROPE of 0.05 *SD*. A determinate verdict on an effect of 0.10 *SD* with probability 0.90 needs 738 distinct texts for the lexical coefficient (640 to 843), 660 for grammatical load (593 to 731) and 634 for continuity (568 to 704). At 0.95, the three need 926, 812 and 784. For a non-negligible verdict, the stricter Bayes-factor criterion under Decision rules needs about 930, 930 and 940 texts at 0.90 and 1,120, 1,120 and 1,130 at 0.95. A component's threshold will be re-derived if the review's pooled estimate for it lies between 0.05 and 0.10 *SD*, and the 0.10 convention is retained below that range, as registered under Design and precision analysis. Below its threshold, a coefficient is declared underpowered before fitting.	The tier-1 Bayesian model specified under Confirmatory models regresses within-language standardised log time per word on the three components, the age band, narration and their interactions. It adjusts for reading level, text length, valence, arousal, narrativity and the generated-or-classic indicator. Text, Language, Country and App version have varying intercepts, the component slopes vary by Language, and an Account intercept enters where the Account rule under Sampling plan is met. The three component posteriors are the registered tests.	Each coefficient is labelled practically non-negligible in the predicted or the opposite direction, practically equivalent to zero, or inconclusive. A combined H1 verdict is recorded only when all three are non-negligible in the predicted direction. A coefficient non-negligible in the opposite direction counts against the corresponding proposition of the theory object. A positive continuity coefficient is not interpreted as a reverse cohesion effect (see Hypothesis 1). Equivalence is evidence that the property's laboratory effect does not reach 0.05 *SD* in this setting. Where the compression diagnostic registered under Confirmatory models falls below its threshold, the narration rule stated with Hypothesis 1 decides whether a component's verdict is taken from its derived unnarrated slope. The same rule sets how a retained pooled verdict is reported. Shown wrong if all three are equivalent: that the lexical, grammatical and discourse determinants of reading established under laboratory control operate on unsupervised reading of machine-generated stories.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Is the association between grammatical load and pace proportionally smaller, or larger, in the older band?	H2. GrammaticalLoad × AgeBand, tested two-sided. The capacity and experience accounts predict a negative interaction, the bottleneck-release account a positive one.	As H1. A determinate verdict on an interaction of 0.10 *SD* needs 1,230 distinct texts (1,107 to 1,359) at an average of twelve readings each for 0.90 and 1,510 (1,349 to 1,680) for 0.95. Declaring equivalence needs about 1,600. The Bayes-factor criterion needs about 1,470 for 0.90, and its 0.95 target was not reached within the 1,600 texts simulated. All these requirements assume a balanced band split and will be multiplied by $1/(4p(1-p))$ at the observed band prevalence before fitting.	The same tier-1 model. The GrammaticalLoad × AgeBand posterior is the registered test.	Negative and outside the ROPE supports the capacity and experience accounts without separating them, and positive and outside supports the bottleneck-release account. Inside the ROPE is evidence that no developmental signature reaches 0.05 *SD* across these two bands, although a non-monotonic trajectory cancelling across two coarse bands, or interactions of opposite sign in orthographies of different depth, would produce the same posterior. The per-language interactions and the finer-age sensitivity analysis, which can show a reversal where the recorded ages are finer than the two bands, are reported with the pooled one. Shown wrong: whichever account or accounts the sign contradicts. A posterior inside the ROPE weighs against all three, since each predicts a signed proportional change.
Is explicit connection between ideas associated with a proportionally larger gain in pace for the younger or the older reader?	H3. ReferentialContinuity × AgeBand, tested two-sided. The knowledge account predicts a positive interaction, the cue-utilisation account a negative one.	As H2. A determinate verdict needs 1,431 distinct texts (1,276 to 1,596) for 0.90, and the 0.95 target was not reached within the 1,600 texts simulated. The probability of correctly declaring equivalence approaches 0.90 at 1,600 texts. The Bayes-factor criterion needs about 1,480 for 0.90, and its 0.95 target was not reached within 1,600. The band-prevalence adjustment applies as for H2.	The same tier-1 model. The ReferentialContinuity × AgeBand posterior is the registered test.	Positive and outside the ROPE supports the knowledge account, on which continuity substitutes for an inference the younger reader finds costly. Negative and outside supports the cue-utilisation account, on which a connective must first be reached and understood before it can help. A lower standard of coherence in the younger band is consistent with the same result, and this design does not separate it from the cue-utilisation account. Inside the ROPE is evidence for neither, with the non-monotonicity caveat of H2 and the further caveat that opposite-signed anchor interactions can cancel within the component, which the registered per-anchor check reports. If the retained component loads below .30 on both connective densities, the sign is interpreted for referential and semantic continuity only. The adjudication between the two accounts is then reported as not addressed. Shown wrong: whichever account the sign contradicts.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Is narration associated with a proportionally larger attenuation of the lexical coefficient than of the continuity coefficient?	H4. The proportional attenuation of the lexical coefficient under narration exceeds that of the continuity coefficient by more than 0.20.	As H1, evaluated only where both narration states occur within the eligible sample. The within-account contrast becomes the primary estimate when the user accounts observed in both states cover at least the adjusted text requirement. ROPE of 0.05 on the difference of proportions. The simulated requirement of 1,234 texts (1,113 to 1,361; 1,506 at 0.95) applies to the lexical-by-narration interaction coefficient and is a lower bound for the difference rule. Under the Bayes-factor criterion, that interaction needs about 1,470 for 0.90. Both requirements will be multiplied by $1/(4p(1-p))$ at the observed narration prevalence, a factor of about 3.0 at $p = 0.91$, and stated before fitting.	Draw-by-draw contrast of the two proportional attenuations, computed from the derived narrated and unnarrated slopes of the tier-1 model as set out under Confirmatory models. Evaluated only if the derived unnarrated slopes of the lexical and continuity components each have fewer than 5 per cent of their draws inside the ROPE (the Hypothesis 4 precondition). Reported with the residual standard-deviation ratio (the compression diagnostic) and repeated in the unpaced subset, the narrated sessions the audio cannot have paced.	At least 95 per cent of the posterior of the difference above 0.20 supports the decoding-scaffolding account, on which narration selectively removes decoding cost. More than 95 per cent inside the ROPE supports the audio-pacing account. More than 95 per cent below -0.05 is consistent either with reverse selectivity, which contradicts both accounts, or with pacing acting through the audio's duration. The duration-adjusted contrast, registered under Sensitivity analyses, distinguishes these two explanations and is reported with this outcome and with the inside-ROPE outcome. Any other result is inconclusive, and both accounts remain open. If the Hypothesis 4 precondition is not met, H4 is reported as not evaluable. Where the compression diagnostic indicates compression, the full-sample verdict is withheld and H4 is evaluated in the unpaced subset. Shown wrong: either the decoding-scaffolding account of narration or the audio-pacing account. The contrast is designed so that the two cannot both be retained as the account of the same readers, and a conclusive verdict indicates whether audio-paced or decoding-scaffolded reading predominates in the narrated sessions.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Do children's own difficulty judgements track the same text properties?	H5. LexicalDifficulty and GrammaticalLoad are associated with higher odds, and ReferentialContinuity with lower odds, of a harder rating, each outside the ROPE, and the three rank by magnitude as in the reading-time model.	Tier-1 readings with a rating. A text rated by fewer than five distinct children does not contribute. The primary model is complete-case, with an inverse-probability sensitivity analysis for non-response. ROPE half-width of log(1.05) on the log-odds scale. At the 30 per cent rating availability seen in the structural sample (the 35 sessions described under Pilot data), the 0.90 target is not reached within the 1,600 texts simulated. The projected requirement is roughly 9,300 to 10,300 generated texts, about 2,700 to 3,000 carrying at least five ratings (Supplementary Section S5). The simulation will be re-run at the observed availability before fitting.	Ordinal cumulative-logit model with the same fixed and grouping structure and language-varying thresholds. The ordering claim is tested by the posterior probability that the three coefficients rank by magnitude as they do for reading time, evaluated against the 0.90 criterion under Decision rules.	All three odds ratios outside the ROPE in the predicted directions, with ordering probability at least 0.90: two measures of the same expended effort, obtained by different routes, converge on the same ordering of text demands. Predicted directions but ordering probability below 0.90: partial convergence, with the rating usable as a difficulty signal but not as a rank measure. Odds ratios inside the ROPE while the reading-time coefficients are non-negligible: the rating is not a usable measure of experienced difficulty, and the ordering is not interpreted. If the older band's odds ratio then lies outside the ROPE, the limit is reported as specific to the younger band and the pooled verdict is retained. Both outcomes null: the components themselves are suspect. If underpowered, the ordering is reported descriptively and convergent validity is recorded as not established. Shown wrong in the third case: that reading time and reported difficulty reflect the same expended effort. Shown wrong in the second: that a child's judgement weights the three sources of demand as reading time does.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Within a child, is more accumulated in-app reading associated with a faster pace?	H6. Within a child, accumulated in-app reading, entered on a log scale and standardised, is negatively associated with log time per word.	Tier 2: accounts with at least three eligible readings of distinct texts, each account taken to be one child. For the unadjusted trend, the simulated requirement is 609 children (528 to 697), each reading five distinct texts, for 0.90 and 731 (621 to 851) for 0.95. The Bayes-factor criterion needs about 810 children for 0.90 and 970 for 0.95. The tier-2 sweep will be re-run at the observed distribution of sessions per child, with the registered covariates added. Below the re-run threshold, the hypothesis is declared underpowered, not untestable.	The tier-1 specification is refitted on tier 2 with a varying intercept for Account. Accumulated words are decomposed into the account's mean and the session's deviation from it. Session order, days since the previous session and days since the account's first eligible session enter as covariates. The deviation term is the registered test.	Negative and outside the ROPE: an association over the observation window, not a practice effect, because skill and practice are reciprocal over longer spans and maturation is the named residual alternative. Positive and outside: contradicts the orthographic-learning account and is interpreted as within-window slowing that the covariates do not capture. Inside the ROPE: evidence against a within-window orthographic-learning signal of at least 0.05 *SD*. The account-mean coefficient is reported with the deviation term but is not a registered test, because the orthographic-learning account and the reverse path from skill to practice both predict it negative. The difference between the two coefficients is reported as a contextual effect, consistent with between-child selection. It is not evidence for selection, because the composition of user accounts by age band, language and entry date also contributes to it, as does the truncation of accumulated reading at the window's opening. Shown wrong: that reading in the app sharpens the word-recognition processes on which reading depends, within an observation window over which the reverse path is checked by the attrition and lagged-pace refits registered with Hypothesis 6.

Hypothesis 1: text properties and reading pace. We predict that, across distinct texts and within reading level and language, LexicalDifficulty and GrammaticalLoad will each be positively associated with log reading time per word, and ReferentialContinuity negatively. The three coefficients are evaluated and labelled separately under the interpretations in Table 1, and the failure of one does not suppress reporting of the others. A positive continuity coefficient is not labelled a reverse cohesion effect, because that effect is a learning outcome (McNamara et al., 1996; McNamara & Kintsch, 1996; O'Reilly & McNamara, 2007). On the construction-integration model, the effect arises when a less cohesive text makes the knowledgeable reader infer more. The extra inference would appear on a time scale as slower reading of that text, in the predicted direction. A grammatical coefficient practically equivalent to zero would extend to children the negligible passage-level effect of global complexity measures reported for adults (Kuperman et al., 2025). That interpretation does not apply to the lexical component, whose anchors are word-level aggregates the adult study did not test at passage level.

The narration rule follows from how narration is distributed. Narration was on in about nine sessions in ten of the structural sample, so each prevalence-weighted main effect is dominated by the narrated slope. If the audio sets the pace of narrated sessions, the narrated continuity slope converges on zero and the lexical slope on the slope of audio duration per word, as explained under Hypothesis 4. The compression diagnostic, the ratio of the narrated to the unnarrated residual standard deviation, signals such pacing when it falls below the cut-off registered under Confirmatory models. Where the diagnostic indicates compression and the distinct texts with unnarrated readings meet a component's threshold, that component's Hypothesis 1 verdict rests on the derived unnarrated slope, with the prevalence-weighted slope also reported. Each component's threshold is set by the seventh adjustment registered under Design and precision analysis. If compression is indicated but the texts with unnarrated readings fall short of that threshold, the prevalence-weighted verdict is retained and reported alongside the unnarrated slope. In that case, an inside-region continuity verdict or a non-negligible lexical verdict is reported as consistent with the audio having set the pace. The rule does not depend on the sign of any coefficient.

Hypothesis 2: grammatical load and the age band. We register the GrammaticalLoad × AgeBand interaction as a two-sided adjudicating test, because developmental accounts predict opposite signs on this study's scale. On the capacity account, a grammatical dependency costs time in proportion to how close its maintenance comes to exhausting the working memory available. As that capacity matures, the same structure costs proportionally less, and the interaction is negative (Just & Carpenter, 1992; King & Just, 1991). Working-memory span does track complex-syntax comprehension and production between the ages of 5 and 12 (Delage & Frauenfelder, 2019). The experience account predicts the same sign but attributes the change with age to growing exposure to complex structure, since exposure makes such structure cheaper (MacDonald & Christiansen, 2002). Evidence that children with more print exposure produce relative clauses more as adults do (Montag & MacDonald, 2015) concerns production, not the cost of reading such a structure.

The bottleneck-release account predicts the opposite sign, locating the change with age in the base over which the proportion is taken and in how fully structure is built on that base. As print-to-sound mapping automatises, the base decoding time shrinks (LaBerge & Samuels, 1974), leaving the reader with representations whose quality makes retrieval cheap (Perfetti, 2007). An unchanged absolute grammatical cost then occupies a larger proportion of a smaller base, and the interaction is

positive. As for how fully structure is built, comprehenders often leave syntactic structure shallow or unresolved, and unresolved structure is read faster, although the evidence for that claim is mixed (Ferreira et al., 2002; Frances, 2024; Swets et al., 2008). Children of 8 to 10 show the same signatures as adults, and the older among them attend more to the punctuation that marks clause boundaries (Wonnacott et al., 2016). On this view, a younger reader who builds shallower structure incurs less of a dependency's cost. The capacity account is the theory object's own proposition, and the bottleneck-release account is registered there as its rival, so a positive interaction would count both against that proposition and for the rival.

The capacity account's premise is less secure in this age range than its formulation suggests. Across 197 studies, the relation between working memory and reading was moderate, $r = .29$, and its contribution to comprehension was accounted for by decoding and vocabulary where those were controlled (Peng et al., 2018). Memory span has been related to relative-clause comprehension in children, though narrowly in both studies we found. In German-speaking 7-year-olds, the relation was to digit span and held for object relatives alone (Arosio et al., 2012), and in French- and German-speaking children, verbal working memory correlated with object-relative comprehension only in French (Solaimani et al., 2025). Partly because the premise may not hold uniformly across the languages in this sample, the interaction is also reported separately by language.

The direct evidence does not decide between the accounts. The largest eye-movement study at these ages, with 81 children reading 12 items, found no attenuation (Wonnacott et al., 2016). It manipulated temporary ambiguity, which indexes reanalysis, a different demand from the sustained maintenance that dependency length imposes, so its null result does not transfer to the present operationalisation. Where an interaction has been reported, it was confined to late measures, in a study with 14 children per group (H. S. S. L. Joseph & Liversedge, 2013). Recomputed from that study's tables as a proportion of baseline, the effects on two of its measures disagree. On go-past time in the target region, our computation gives 15 per cent in the younger children, 25 per cent in the older and 31 per cent in adults. That rise with age is the pattern a positive interaction would produce on a proportional scale. On post-target total reading time, where the study's own significant interaction lay, the proportional effect ran the other way. It was 27 per cent in the younger children, 12 per cent in the older and 4.5 per cent in adults.

A third possibility is that development across these ages is not monotonic at all. With 250 children aged 8 to 11 reading and hearing sentences word by word, the largest existing test reported a U-shaped developmental pattern in the online processing of restrictive relative clauses (Booth et al., 2000). Offline comprehension of complex syntax grows most steeply between 5;6 and 7 years (Frizelle et al., 2025), so the younger band falls immediately after the steepest phase. The boundary between the bands coincides with a documented shift in online processing. In a study of Russian-speaking children (Kruchinina et al., 2026), accuracy on passive sentences rose sharply between 7 and 8. Although the 8- to 10-year-olds then matched adults on accuracy, the incremental-revision signature at the critical word, present in adults and at 8 to 10, was absent at 6 to 7. Accuracy converged before processing did, and that difference shows in time-based outcomes.

Table 1 interprets each Hypothesis 2 outcome and states two caveats on a posterior inside the region.

Hypothesis 3: referential continuity and the age band. We register the ReferentialContinuity × AgeBand interaction as a second two-sided adjudicating test, because theory supplies two moderators

of coherence facilitation that push the interaction in opposite directions. They are background knowledge and knowledge of the cue, each rising between 7 and 11. The construction-integration model (Kintsch, 1988), together with what exploiting a cohesive cue requires, supplies both. On the knowledge account, a gap in the text is bridged from knowledge that the younger reader has less of and retrieves less readily. Age differences in coherence inferencing between 6 and 15 persisted when the relevant knowledge was made equally available to every child, and its accessibility mattered more for younger children (Barnes et al., 1996). At 11 to 12, higher knowledge went with faster reading of sentences that required an inference (Cain et al., 2024). Surface continuity substitutes for an inference that is costlier at 7 to 8 than at 9 and over, so it is worth more to the younger reader and the interaction is positive.

On the cue-utilisation account, exploiting a connective requires enough decoding fluency to reach it and enough explicit knowledge of what it signals, so the older reader gains more and the interaction is negative. Knowledge of causal and adversative conjunctions is still developing between 8 and 10 (Cain et al., 2005; Cain & Nash, 2011), and comprehension of connectives grows across the primary years (Volodina & Weinert, 2020). Like the capacity account, the knowledge account is the theory object's own proposition, with the cue-utilisation account registered there as its rival, so a negative interaction would count against that proposition.

The cue-utilisation account depends on decoding fluency, so it implies a larger negative interaction where decoding is still maturing across the two bands, which the orthographic-depth paragraph below places in English, French and Portuguese. The knowledge account, resting on background knowledge and narrative experience, implies no such dependence. The per-language interactions registered under Sensitivity analyses are reported partly for that reason.

Decoding fluency and knowledge of the cue develop well past the ages studied here. Knowledge of what a connective signals is still incomplete in French-speaking teenagers of 12 to 19 and native Russian-speaking teenagers of 11 to 17. In both samples, vocabulary size and exposure to print were stronger correlates of appropriate use than age (Tskhovrebova et al., 2022, 2023). Both moderators are likely to grow with the reading a child accumulates, a further reason the net sign of the interaction cannot be derived in advance.

The two accounts concern different anchors of ReferentialContinuity. On the knowledge account, referential overlap and semantic similarity between adjacent sentences spare the reader a bridging inference, whereas the cue-utilisation account relies on connectives. In adults, semantic and causal relatedness dissociate by reader skill, with semantic relatedness weighing more for the less skilled reader (Todaro et al., 2010). They also bear differently on processing time and on what is remembered (Wolfe et al., 2005). The anchors need not therefore show the same age interaction, and a component interaction inside the region could arise from anchor interactions of opposite sign. Where the component posterior lies inside the region while the anchor interactions registered under Sensitivity analyses lie outside it with opposite signs, the equivalence verdict is retained and reported with that pattern. The sign of a coherence effect also depends on the cohesive device involved, so connectives enter the component by relation type, and the causal and contrastive densities are its connective anchors. The facilitation a connective affords was smaller for result relations than for concessive ones and differed between an analytic and a synthetic language (Marchal et al., 2025), which supports both halves of that decision.

A negative interaction has a second interpretation under the Landscape model, which makes a reader's attempt at a bridging inference depend on the standard of coherence they bring to the text (van den Broek et al., 2005; van den Broek & Helder, 2017). A reader who does not attempt it pays no time for the gap, so continuity would save the younger reader nothing if that band read to a lower standard. The online evidence limits that concern, since children of 8 to 13 made the local inferences a coherent representation required, if not the optional ones (H. Joseph et al., 2021). This design does not separate a lower standard of coherence from cue utilisation, so Table 1 treats a negative verdict as consistent with both.

As with Hypothesis 2, the existing evidence does not decide between the accounts. Eight- and ten-year-olds gained the same speed-up from an appropriate connective in self-paced reading, with the age interaction explicitly null across 91 children (Cain & Nash, 2011). No difference in online sensitivity to a coherence break was found between children of 8 to 9 and of 10 to 11 (Helder et al., 2016). Among 8- to 10-year-olds reading two-clause sentences, the absence of the Dutch causal connective lengthened processing only for second-language readers with weaker syntactic knowledge (Van den Bosch et al., 2018). The connective was worth most to the reader with the fewest resources, but the contrast between language backgrounds leaves age untested. In fourth graders, the effect of cohesion turned on world knowledge and far less on decoding skill, with age untestable within a single grade (McNamara et al., 2011). Where the studies we found at these ages manipulated whole-text coherence, they measured comprehension or inference, so none bears directly on reading rate. The three outcomes are interpreted as for Hypothesis 2.

The bottleneck-release account links the two age-band interactions. The smaller base that raises the proportional grammatical cost in the older band also enlarges the proportional continuity saving there. The account predicts a positive Hypothesis 2 interaction, a negative Hypothesis 3 interaction and a negative or null LexicalDifficulty × AgeBand interaction. We will report the joint pattern of these three tier-1 estimates descriptively alongside the two registered coefficients, and no verdict depends on it.

Both age-band interactions are averaged over ten languages whose orthographies differ in depth, and the age band does not mean the same thing in each. In most European orthographies, children read familiar words accurately and fluently within the first school year, whereas development is more than twice as slow in English and lags in French and Portuguese (Seymour et al., 2003). Psycholinguistic grain size theory derives that pattern from how consistently a writing system represents phonology, which sets the size of the units a learner comes to rely on (Ziegler & Goswami, 2005). Children matched on years of instruction show orthography-specific length and frequency signatures, with English qualitatively unlike German and Finnish, while adults converge (Schroeder et al., 2022). Between two transparent orthographies, the developmental trajectory can nonetheless be comparable, with language-specific effects confined to morphological complexity (Marzi et al., 2026), so the heterogeneity expected here is a matter of degree. Children reading different orthographies are therefore at different points on the automatisation curve. On the bottleneck-release account, the developmental signature would be largest where decoding automatises latest. The registered coefficient is a prevalence-weighted average across those positions, and we do not generalise it from any one orthography (Share, 2008).

Hypothesis 4: narration and selective attenuation. We predict that the LexicalDifficulty coefficient will be proportionally smaller under narration than the ReferentialContinuity coefficient is. With

π_{lex} and π_{cont} denoting each coefficient's proportional attenuation under narration, computed draw by draw as set out under Confirmatory models, the prediction is that $\pi_{\text{lex}} - \pi_{\text{cont}} > 0.20$. The threshold is four times the region's half-width and is a registered convention. The prediction follows from the decoding-scaffolding account. A spoken rendition supplies word identity without print-to-sound mapping, the step that a low-frequency or long word makes expensive. The rendition does not supply the link between two ideas, which the reader must still construct. The benefit of text-to-speech is largest for children whose decoding is weak but whose language is intact (Keelor et al., 2023). In 34 children aged 8 to 9, the time saved by the audio fell on the first encounter with an orthographically novel word and had gone by the second (A. Valentini et al., 2024). Besides word identity, a voice conveys prosodic phrasing, which a silent reader must project (Breen, 2014) and which is still developing across the primary years (Godde et al., 2020). Narration could thereby lower the grammatical coefficient by a route unrelated to word identity. Prosodic phrasing does not, however, supply the link between two sentences, so the contrast between the lexical and continuity attenuations remains interpretable. The grammatical-by-narration interaction is reported for description only.

Whether narration spares the reader the decoding cost depends on timing. In reading while listening, adults' eyes generally ran ahead of the audio and rarely rested on the word being heard (Conklin et al., 2020). Among 119 learners of English, the multimodal condition ran slightly slower than reading alone (Malone et al., 2025), whereas children of 8 to 9 showed the opposite pattern (A. Valentini et al., 2024). The decoding account describes a reader slower than the audio, and the competing audio-pacing account a reader faster than it. The full-sample contrast shows which kind of reader predominates in the narrated sessions.

On the pacing account, the audio's duration sets narrated reading time, so each narrated slope converges on the slope of audio duration per word. That slope is near zero for referential continuity and positive for lexical properties tied to word length, since a word with more syllables takes longer to speak. The two attenuations are therefore equal only once that duration is accounted for, and otherwise the pacing account predicts $\pi_{\text{lex}} - \pi_{\text{cont}}$ at or below zero. Reading aloud runs at about 183 words per minute, against 238 to 260 for silent reading, and the gap between text types is predicted by word length (Brysbaert, 2019). Pacing alone fails to flatten word-level effects, since manipulating reading speed left the influence of word properties on eye movements in place (Schwalm et al., 2026). That result partly motivates the registered contrast between the lexical and continuity attenuations.

The compression diagnostic withholds the full-sample verdict where the audio sets the pace for most narrated sessions. An adjustment for the audio's duration allows for the mixture that the diagnostic can miss, in which only some narrated sessions are paced. The adjustment rests on a predicted spoken duration per word, defined under Materials and computed from the texts before any outcome is linked to them. The refit including it is registered under Sensitivity analyses.

The pacing account has substantial support. The meta-analytic comprehension benefit reported above is confined to comparisons with experimenter-paced reading (Clinton-Lisell, 2023). A registered report testing the resource-freeing mechanism in 86 adult second-language readers found comprehension *worse* under narration than in silent reading (Hui & Godfroid, 2026). In the same study, individual differences in orthographic decoding and speech segmentation did not interact with input condition in the way that mechanism predicts. In 21 secondary-school students with dyslexia and 22 without, audio support increased reading time in most tasks while leaving comprehension

unchanged (Knoop-van Campen et al., 2021). Where measured incidentally, the compression is severe, with the standard deviation of exposure time falling from 11.56 seconds when self-paced to 0.62 seconds when narrated (A. Valentini et al., 2024). The tests we found concern comprehension and the variance of exposure time, and none bears on the contrast that Hypothesis 4 registers.

A third possibility, with the same predictions as the pacing account, is that the child listens without reading, so that the interval is the audio's duration plus the page turns. Attention to print is low in preschoolers listening to a storybook (Evans & Saint-Aubin, 2005) and rises across the primary grades with reading skill (Roy-Charland et al., 2007). In older children reading an illustrated story in a second language, audio moved the eyes from the text to the pictures without changing comprehension (Pellicer-Sánchez et al., 2020; Serrano & Pellicer-Sánchez, 2022). By contrast, 8- to 9-year-olds did fixate the words under bimodal presentation (A. Valentini et al., 2024). The share of narrated sessions in which the child read along is unknown and likely to differ by age band. The logs cannot separate a paced reading from a listening. An inside-region verdict is therefore reported as time set by the audio, whether or not the child read along, and never as evidence that the narrated sessions involved reading. Listening without reading is ruled out by definition only in the narrated sessions that the audio cannot have paced, the unpaced subset defined under Confirmatory models.

The Hypothesis 4 contrast is exposed to selection into narration state. With narration on in about nine sessions in ten, the unnarrated slopes, the denominators of both attenuation proportions, are estimated largely from the families who chose to switch the audio off. If the children reading without audio are disproportionately the fluent readers, their slopes are attenuated by selection and both proportions are distorted. The data contain no measure of reading skill with which to adjust for this, so we will report narration prevalence by age band, reading level and language before fitting. The rule under Confirmatory models makes the within-account contrast the primary estimate where the data allow. Because both proportions are ratios, the contrast is evaluated only under the Hypothesis 4 precondition, that fewer than 5 per cent of the draws of each unnarrated slope, lexical and continuity, lie inside the region. If either slope fails the precondition, there is no attenuation to compare and Hypothesis 4 is reported as not evaluable. The precondition does not depend on the sign of any coefficient.

The four outcomes, their posterior criteria and their interpretations are registered in Table 1 and under Decision rules.

Hypothesis 5: text properties and the difficulty rating. We predict that, in an ordinal model of the post-reading rating, LexicalDifficulty and GrammaticalLoad will be associated with higher odds of a harder rating and ReferentialContinuity with lower odds, each by an odds ratio outside the region. We further predict that the three standardised coefficients will rank by magnitude as they do in the reading-time model. Two measures of the same three constructs, one behavioural and one judgement-based, obtained on different scales from overlapping sets of children, need not rank those constructs the same way. Their agreement on order is therefore a substantive prediction of the theory object, on which both measures record the same expended effort.

The link between the two measures rests on the effort heuristic of Koriat's cue-utilisation approach to metacognitive judgement. On that approach, a judgement of difficulty is inferred from cues such as experienced effort and does not directly reflect processing (Koriat, 1997; Undorf & Erdfelder, 2011). Two meta-analyses put the association between experienced effort and monitoring judgements at *r*

= -.36 and, in a meta-analytic structural model, at β = -.19 (Baars et al., 2020; David et al., 2024). Those are judgements of learning and of confidence, so for a difficulty rating the sign is reversed. Second, third and fifth graders were less confident in an answer the longer they had taken to choose it, and reliance on that cue was stronger in the older children (Koriat & Ackerman, 2010). The rating would then be a later report of the same expended effort that reading time measures, not a separate measurement of the text. Effort is attributed to the material when the material imposes it, but not when a learner chooses to invest it, a distinction established in adult learners (Koriat et al., 2006; Koriat et al., 2014). In children, consistent with that distinction, the association between effort and judgement weakens where regulation is goal driven (Baars et al., 2020).

Partly because reading in this app is self-paced and voluntary, the rating is registered here as a measure under test. On the same grounds, the theory object marks this prediction as diagnostic against the engagement account. Convergence in the predicted directions weighs against that account. Its absence is interpreted, under the rule in Table 1, as a failure of the rating and not as support for the engagement account.

That approach also suggests why the two measures might diverge. A judgement made after finishing draws on the cues still accessible at that moment, and the three sources of demand are not equally accessible. Word-level difficulty recurs at every line, whereas the cost of an unstated link between ideas is spread over the text. Consistent with this, a cohesion manipulation that moved paragraph-level judgements left whole-text judgements untouched unless readers could still consult the text (Lefèvre & Lories, 2004). In adults predicting their own recall, however, judgements about a whole text did rise with coherence (Rawson & Dunlosky, 2002). The posterior probability that the two measures rank the components alike is the study's principal convergent-validity check, evaluated against the 0.90 criterion under Decision rules.

The evidence closest to this rating concerns related judgements, mostly those children make about their own learning and comprehension. Among children of 8 to 10 solving arithmetic problems, the mental effort they reported investing was related to their judgements of learning, which tracked the complexity of the problem (Baars et al., 2018). Children in both of this study's age bands judge how well they have understood a text (Steiner et al., 2020), and fifth graders do so both for the whole text and for individual answers (Markovich et al., 2026). Primary-school children are more overconfident in those judgements than secondary-school children (Mirandola et al., 2018), but that overconfidence depends on the text. Among 407 primary and secondary students, the primary children overestimated their performance on expository passages and inferential questions but calibrated as well as the older students on narrative texts (Zagato et al., 2026). Every text in this study is narrative. Comprehension judgements concern the reader's own understanding, whereas the rating here asks about the text. In adults, subjective comprehensibility ratings of narrative text can be predicted from computed text features (Tilmatine et al., 2024). None of the child metacognition literature we found validates a global post-reading judgement of a text's difficulty against the text's measured properties in children of this age. We likewise found no psychometric evaluation of the three-point format, a classroom heuristic whose options are easy, just right and difficult.

Children's monitoring judgements are internally consistent when aggregated over items, their reliability is limited by that of the first-order task, and their accuracy varies from one task to another (Eberhart et al., 2025; Roebers et al., 2021). The design therefore aggregates ratings at text level and applies the five-rater floor registered under Confirmatory models. Children of 6 to 13 have

a limited grasp of multi-point response formats, and word-anchored options agree better with a yes-or-no standard than numeric ones do (Mellor & Moore, 2014). The number of options bears on how reliably children and young adolescents answer, with four options proving optimal and seven or more lowering reliability (Borgers et al., 2004). Which way any of this would move a three-point rating inside an app is not established, so we register the direction as unknown and report the observed distribution. Equating reported difficulty with expended effort is contestable. In 80 children aged 8 to 12, the difficulty and effort ratings given after a battery of cognitive tasks were uncorrelated with each other and only weakly related to performance (Torres & Toplak, 2026).

With age, children come to judge a task by the effort it cost. In self-paced learning, judgements of learning fell as study time rose for third to sixth graders, aged 9 to 12, but not for first and second graders, aged 7 and 8 (Koriat et al., 2009). That division coincides with this study's two age bands but does not show whether direct judgements of a text divide the same way. A rating that tracks the text at 7 to 8 remains possible, because even 5- and 6-year-olds use an item's difficulty adaptively when evaluating their own performance (van Loon et al., 2017). The rating model includes the same component-by-age-band terms as the reading-time model, but its registered tests remain the pooled odds ratios and the ordering. The band-specific odds ratios derived from those terms are descriptive quantities, as stated under Confirmatory models. The recorded age bands in the structural sample skewed towards the younger of this study's two bands (Supplementary Section S8), so a prevalence-weighted pooled odds ratio will be weighted towards that band. Table 1 accordingly interprets a pooled odds ratio inside the region, together with an older-band odds ratio outside it, as a limit on the rating specific to the younger band.

The effective sample size is the number of distinct children rating each text, which the raw count of ratings overstates, so the model has a varying intercept for Text. We will report the intraclass correlation for the rating at text level as a reliability estimate for this format. Before fitting, we will report the distribution of raters per text and the distribution of the three rating categories overall and by age band and language. The category distribution will be reported again alongside any inside-region verdict, because a distribution concentrated in one category limits what a cumulative logit can resolve.

A failure here is ambiguous on its own, since it could lie in the rating or in the components. Because Hypothesis 1 resolves the ambiguity, the interpretations in Table 1 are conditioned on the reading-time coefficients. Components that predict reading time but not the rating leave the feature pipeline sound. A mismatch in ordering alone is reported as evidence about which cues children use and does not call the components into question. The design analysis marks this hypothesis as the one most likely to be underpowered. If it is declared underpowered, or its verdict is inconclusive, the ordering is reported descriptively and convergent validity is recorded as not established. This rule gives the fifth outcome-neutral check a defined outcome whether or not Hypothesis 5 reaches a verdict. An inconclusive verdict from the census of existing records is still informative, because it sets an upper limit on the reliability a single-item difficulty rating can have at the availability such apps achieve.

Hypothesis 6: accumulated reading and reading pace. We predict that, among children with several recorded sessions, a one-standard-deviation within-child increase in accumulated in-app reading, on its log scale, will be associated with a reduction in log reading time per word. The reduction is predicted over and above session order, the interval since the previous session and the time since the child's first session. The unit is the standard deviation because the region is defined in that unit

under Decision rules. The implied change per doubling of accumulated words will be reported with the coefficient.

On the orthographic-learning account, each successful encounter with a written word form is an opportunity to acquire or sharpen the word-specific orthographic representation supporting its later recognition (Share, 1995). Under the Lexical Quality Hypothesis, the quality of those representations makes skilled word reading cheap (Perfetti, 2007). Print experience is accordingly treated as the main source of growth in word reading (Castles et al., 2018; Deacon et al., 2024; Nation, 2017). Because the mechanism operates on items, exposure is counted in words of completed distinct stories, and repeated readings are reserved for the practice analysis under Sensitivity analyses. Each further form adds less as the lexicon fills, so accumulated reading enters on a log scale.

Learning of this kind is primarily item-based. In grades 1 and 2, below this study's age bands, children who met novel spellings in short stories learned them, but transferred that learning clearly only to orthographically related forms (Tucker & Deacon, 2025). Across 62 experiments, the outcome of such learning is modulated by language-specific word properties (Li & Wang, 2023). Reading one story can make the next cheaper by two routes, the words the two texts share and the sublexical regularities a reader generalises from them (Castles et al., 2018). The size of the association is therefore limited by how much of a new text earlier reading can already have made easier. Both limits are looser for stories written to one reading level than for texts sampled at large, and neither is measured here.

Outside English, orthographic learning, lexical knowledge, reading fluency and spelling were associated in German-speaking primary-school children (Schmalz et al., 2025). The account's implications differ between the ten languages, because decoding automatises at very different ages across them, as the paragraph on orthographic depth above records (Seymour et al., 2003). On a model in which decipherability precedes unitisation and automatisation (Share, 2025), the per-unit speed-up that accumulated reading affords should be shallower where decoding was automatised earlier. The pooled deviation coefficient remains the registered test. The per-language slopes from the secondary refit under Sensitivity analyses are reported descriptively alongside it and annotated with orthographic depth for the eight sampled languages Seymour and colleagues covered. Polish and Korean are reported unannotated.

As noted earlier, practice and skill are reciprocal over years (Torppa et al., 2020; van Bergen et al., 2021), so the coefficient estimates a within-window association, not a practice effect. Random-intercept cross-lagged models on more than 5,100 German secondary students found no within-person evidence that reading more yields better reading competence, while the reverse path, from competence to exposure, remained (Bittmann, 2025). Only a design that separates within-person change from stable between-person differences can show such an asymmetry. The within-child deviation and the account-mean term used here separate them by definition. For the path from reading to skill, exposure to a text experimentally speeded skilled adults' later recognition of the low-frequency words in it (Chetail, 2024). That study, the closest experimental demonstration we found, is confined to adults and to rare words. Reciprocity established at yearly lags does not by itself license treating the reverse path as negligible over a window of months.

Within the window, a child whose session went slowly may not return, so later sessions would over-represent children whose earlier sessions went well, which would push the coefficient in the

predicted direction. The family of a child reading faster may choose longer stories or raise the reading level. Either change alters the composition of what accumulates, even though the current story's length and level are adjusted for. Before the coefficient is interpreted, we will report the distribution of each user account's final-session pace relative to that account's own mean, and the number of accounts whose reading level changed within the window. The audit under Sampling plan establishes that number in any case. Two refits will be reported with the registered term, and no verdict depends on either. One will exclude every account's final eligible session, and the other will add the previous session's standardised pace as a covariate, by analogy with the random-intercept cross-lagged model (Hamaker et al., 2015). Because conditioning on a lagged outcome alongside an Account intercept is biased in short series, the lagged-pace refit constrains the size of the concern without resolving it.

The account-mean term is reported, but it tests neither the orthographic-learning account nor its reverse-path rival. Both predict it negative, since a child who has read more in the app may read faster for having done so and a child who reads faster may go on reading. How much and how well children read have long been associated at the between-person level (Mol & Bus, 2011). Only the within-child deviation separates the two accounts, and the rival is substantive. In three twin samples, the modelled direction of causation ran from reading ability, literacy skill or comprehension to print exposure or literacy enjoyment, with the reverse influence zero in one sample (Erbeli et al., 2020; van Bergen et al., 2018; van Bergen et al., 2023).

Each covariate addresses a named alternative. Session order absorbs familiarity with the interface and with the generator's story schemas, since a reader who knows the shape of a story reads it faster for reasons the orthographic-learning account does not claim. Days since the previous session absorbs spacing, and days since the first session absorbs maturation and schooling. Once the three are held constant, the accumulated-words deviation is identified by within-child variation in the lengths of the stories already read. That variation is small where the reading level fixes story length. Partly for that reason, Design and precision analysis re-derives the tier-2 requirement at the observed distribution before fitting. Oral reading fluency grows appreciably and unevenly over a school year (Christ et al., 2010), so an elapsed-time covariate absorbs maturation only if growth is close to linear in days within a child and the window's position in the school year is immaterial. Growth rates differ between children, and the variables that predict those rates (Khanolainen et al., 2024) predict how much a child reads as well. Before Hypothesis 6 is interpreted, we will report the within-child correlation matrix of the four covariates and the deviation term, with their variance inflation factors. At the same stage, we will report the calendar month of each session as a descriptive stratifier, to identify any window spanning a school holiday. For the deviation term, Table 1 registers the interpretation of a negative, a positive and an inside-region posterior.

Every hypothesis is labelled by the rule under Decision rules, which pairs a posterior criterion with an interval Bayes factor, and an inconclusive result is reported as such. No hypothesis is filtered by a preliminary significance test, but three labels take precedence over a verdict. A hypothesis is declared untestable if its component fails the construction rule under Outcome-blind dimension reduction or if its outcome or contrast is absent from the frozen extract. That label is assigned before the model is fitted and without reference to any coefficient. A hypothesis whose data fall below its precision threshold in Table 1, as re-derived where Design and precision analysis says so, is instead declared underpowered. Its model is still fitted and reported, and no conclusion is drawn from

an interval the design analysis predicted would be indeterminate. A hypothesis is reported as not evaluable when a registered post-fitting contingency removes its test. Only two contingencies do so: failure of the Hypothesis 4 precondition and failure of the positive control on recorded reading time. Where the data controller's answers to the questions in Table 2 falsify an auxiliary assumption, the affected predictions are reassigned by the rules in that table, never reinterpreted after the fact.

Whatever the verdicts, this study will not identify a causal effect of any text property, age band or narration setting. Nor will it estimate how fast children in general read, or generalise beyond families who chose to install and use this application. Its confirmatory results will say nothing about comprehension, since none of the contingent hypotheses is activated at submission. The constraints on generality are these: children aged 7 and over who read narrative texts of a few hundred words on a touchscreen, in one of ten languages, inside one commercial application, whose generated stories one generator produced (Simons et al., 2017).

The medium is one of those constraints, and it is not neutral. Reading on screen yields poorer comprehension than reading the same text on paper, with the shortfall larger under time pressure and for informational text (Delgado et al., 2018). In children, the paper advantage holds except where a digital text adds features aligned with the content (Furenes et al., 2021). A network meta-analysis of 56 samples locates most of that shortfall in scrolling (Clinton-Lisell & Litzinger, 2026). Where scrolling was required, paper outperformed every device by 0.35 to 0.48 standard deviations. Where it was not, the differences ran from 0.03 to 0.12 and were unreliable, and a tablet read without scrolling ranked closest to paper (Clinton-Lisell & Litzinger, 2026). This app presents a story as pages on a tablet, so the largest known moderator of the screen shortfall does not apply here. The laboratory results underlying the three components were obtained largely on paper, so the comparison drawn here spans a change of medium as well as of setting. A coefficient smaller than its laboratory counterpart may reflect either change.

Methods

Ethics information

This research complies with all relevant ethical regulations. The data are generated in the ordinary operation of a commercial application and collected by its publisher under the terms of use and privacy policy a family accepts on installation. A parent or guardian registers consent in the app for research use of anonymised records. That registration is the informed consent under which the records are transferred to us, and no further consent is sought. We play no part in collection, have no contact with any child or family and return nothing to the platform, so nothing done here can alter what any child experiences. No compensation is provided or applicable.

The extract contains no direct identifier. It records no name, contact detail, precise location or device identifier, and accounts appear in it only as pseudonymous strings generated by the platform. Ethical approval for secondary analysis of these anonymised records [has been granted by / is being sought from] the Departmental Research Ethics Committee of the Department of Education, University of Oxford (reference [to be inserted]). Approval will be in place before in-principle acceptance. The study involves no experimental manipulation or intervention, so no question of withdrawal from a procedure arises. A family's right to have its records erased is exercised through the platform. Any erasure between building the extract and transferring it will be recorded by the data controller and reported at Stage 2.

Pilot data

The pilot material comprises the data dictionary, a 35-row structural sample of the session table and a summary of that sample computed with the count script described under Data provenance. We received the sample and the dictionary from the data controller on 27 July 2026 to confirm the shape of the table, and computed the summary on 6 August 2026. The sample contains real reading times for 35 sessions on five distinct texts, one session per account, in ten reading languages. It cannot be used to test any hypothesis, since none is identified by 35 observations and no within-child term is estimable from one session per account. The sample nevertheless fixed three inputs to the design analysis: the narration prevalence (on in 32 of 35), the rating availability (present in 10 of 35) and the age-band missingness (blank in 10 of 35). Supplementary Section S8 reports the sample and its summary in full, and the sample is deposited for peer review as the Data availability statement describes.

Design

Data provenance, sequestration and outcome blindness

The data this protocol analyses already exist. Nature Human Behaviour considers secondary analyses of existing data on condition that authors can evidence no prior access to the data in question. We put the arrangement set out below to the journal's editorial office, which confirmed it as appropriate on 3 August 2026.

The data controller builds the extract on its own schedule and holds it. No part of the extract will be transferred to us until the journal confirms in-principle acceptance in writing. Until then, and with the editorial office's agreement, three items may be sent to us: a manifest, the story texts and the provider-computed counts. The manifest gives the column list, the row count, the cut-off timestamp, the build date and a SHA-256 checksum for every file. With it, the file eventually transferred can

be shown to be the one that was built. The story texts are identified by story identifier, reading language, reading level and app version, and their page breaks are preserved. They may be received before in-principle acceptance because every linguistic measure is computed on them and none contains a behavioural outcome.

The provider-computed counts are aggregates that the data controller obtains from the extract by running a count script we supplied. The count script reads nineteen columns besides reading time, writes one small table of counts and a one-page summary drawn from it, and never reports a reading time as a value. Both the version sent on 7 August 2026 and a revised version are deposited with this protocol. The revised version, sent on [date], adds checks against malformed input without changing what is counted, and it is the version the data controller will run.

A count computed from an outcome column could compromise blindness, so the script treats reading time only as a column whose usability has to be established. It reports how many values are absent, non-numeric, negative or zero, how many fall into each of seven bands bounded by successive powers of ten, and how many distinct values the column contains. The script also reports the largest number of sessions sharing any single value, and it never crosses reading time with language, age, text, account or date. Supplementary Table 5 shows its output on the structural sample. Bands this coarse cannot distinguish a fast text from a slow one, but they do distinguish a column recorded in seconds from one recorded in milliseconds. The analysis cannot be specified until the column's unit is known. A session-timeout cap would appear as a large number of sessions sharing one value, and the censoring rule under Sampling plan depends on that count.

At submission, we hold the story texts, received on [date]. The manifest and the provider-computed counts [had been received on DATE / had not yet been received]. The number of distinct texts with eligible readings cannot be established without those counts and is reported in Table 2 as unresolved.

Two conventions apply throughout this protocol. First, whether a hypothesis is tested, on which sample and with which model is decided from the provider-computed counts, the data controller's written answers and the thresholds registered here. No fitted quantity enters these decisions except through the post-fitting rules listed under Decision rules. Second, wherever the text says a quantity will be reported, it will be reported in the Stage 2 manuscript whether or not it changes a conclusion.

A descriptive study of an earlier 30-day window on this platform, published by two of the authors, used only summaries the data controller prepared (Ratner & Murphy, 2026). That study, described in the Introduction, relates no text property to any behaviour, analyses no reading duration and informs no decision registered here. The structural sample and its summary are disclosed under Pilot data.

At the editorial office's direction, we have not run the dimension reduction specified below and will not run it before submission, although we hold the texts on which it would operate. The component solution is fully specified but unfitted.

The data controller will supply a signed copy of the certification reproduced in Supplementary Section S7. The document states the held extract's build date, window, row count and checksums and names its owner and a deputy. It certifies that no part of the extract has been or will be transferred to the research team, or to anyone acting for it, before written confirmation of in-principle acceptance. A final clause states that no analysis relating story text to reading time has been conducted by the

data controller or reported to the team.

Materials

The reading materials are the story texts, each written by a generative model around the combination of five attributes a family chooses. Once generated, a text is fixed and is not rewritten when the same combination is chosen again. Families may also read from a fixed classic set of texts. For each reading, the extract records whether the text was generated or classic, the reading level served, the language, the app version and the page count. Every session will be linked to the features of the exact version it was served, matched on story identifier, reading language, reading level and app version.

Every text coefficient is identified by comparing texts. The number of distinct texts with eligible readings is therefore the single quantity that governs how much this design can establish, and it is not yet known. The data controller's estimate spans a few hundred to a few million, depending on whether one combination of chosen attributes yields one shared text or a separate text per session. The platform's published description provides the working assumption on this question and on how texts differ across languages. It describes a catalogue of over 40,000 English-language stories generated from attribute combinations, many of them translated into the child's home language, and over 2.4 million story sessions in the first six months after launch (Ratner et al., 2025). If that description holds for the window, texts are shared across sessions and differ across languages by translation. The models would then include the varying intercept for the underlying story, crossed with Language, that Table 2 registers under question (d). The data controller's written answer supersedes that assumption, and the same table gives the decision rule for each answer.

For every text, we will compute features in four families. *Lexical* features are mean log frequency of content words, the proportion of words with a frequency below one per million tokens on the uniform frequency resource, mean rated age of acquisition and mean orthographic length. Mean per-word surprisal under a language-specific neural language model, lexical density, lexical diversity and concreteness complete the family. Diversity is the moving-average type-token ratio over a 50-token window, and its per-language correlation with the Measure of Textual Lexical Diversity will be reported. A word's surprisal is the sum of the surprisals of the subword tokens composing it, and the mean is taken over the same whitespace-delimited words as the outcome's denominator. *Grammatical* features are mean dependency distance from a universal-dependencies parse, clauses per sentence, the ratio of subordinate to main clauses, mean sentence length and mean depth of noun-phrase modification. *Discourse* features are mean semantic similarity between adjacent sentences from a multilingual sentence encoder, referential overlap, connectives per hundred words by relation type and narrativity. Referential overlap is the proportion of adjacent sentence pairs sharing at least one content lemma in a nominal position, which operationalises the argument-overlap rule of the construction-integration model without a coreference resolver. Narrativity is a stipulated composite of past-tense marking, personal pronouns and direct speech, defined in Supplementary Table 3. *Affective* features are the mean and the variability of valence and of arousal. One further text quantity serves Hypothesis 4 alone, a predicted spoken duration per word on a log scale, computed from syllables per word plus sentence- and clause-boundary pauses per word from punctuation. Its resources and matched-token proportion are given in Supplementary Table 3. Where the data controller supplies the per-text audio duration, that duration replaces the prediction, under the rule given below.

The extract records narration only as a session-level flag, and the platform has not disclosed how the audio is produced or presented. Two undisclosed properties bear on the interpretation of a narrated session. Word-level highlighting couples the eye to the voice, although the evidence comes from children younger than any here (Liao et al., 2020). A page that advances when playback ends would make a narrated session's duration largely the audio's duration. Question (a) in Table 2 lists these and the remaining items with the consequence of each answer. Any item established from the public version of the app is reported as observed, with the date. Observing the app involves no outcome data, and none of these facts enters a model.

Supplementary Table 3 gives the primary and the sensitivity resource for each feature family, with its coverage across the ten languages the structural sample showed. Four rules govern the table and are fixed before any feature is computed. First, each feature will be computed from a single resource applied identically in every language, even where a better native resource exists in one of them. Native norms will be run as a pre-specified sensitivity analysis, and their correlation with the single-resource values will be reported for each language. Second, no value is transferred across languages by translating a word list. A predicted resource is admitted only where the prediction was made within each language by one documented procedure and its correlation with native ratings is reported. Where no acceptable measure exists for a language, that language does not contribute that feature.

Third, every measure reports the proportion of tokens it matched, and a language in which a measure matches fewer than 80 per cent of content-word tokens does not contribute it. Content-word tokens are those whose universal part-of-speech tag is NOUN, VERB, ADJ or ADV. The token-coverage rule differs from the text-coverage rule under Outcome-blind dimension reduction, and the two can reach opposite decisions on the same feature. Where some languages lack a feature, that family's pooled solution is computed on the features common to every contributing language. Such a feature enters only the per-language descriptive solutions and the language-specific sensitivity analyses.

Fourth, the full inventory, the software versions, the model checkpoints, the tokenisation rules and the matched-token proportions will be frozen and published before any feature is linked to outcome data. Sentence segmentation is registered with them: pages are concatenated with a single space, and a page break is never a sentence boundary. Supplementary Section S3 lists the per-language segmentation diagnostics to be reported. The four rules constrain Korean most, and the same section sets out the consequences for it after Supplementary Table 3.

Proper nouns are excluded from the content-word denominator of the third rule and from every norm-based mean, because the platform puts invented names into every story. The hero, the companion and the setting a family chose are named in the text, and no subtitle frequency list, age-of-acquisition norm or concreteness norm contains an invented name. Those names follow from the family's choices and recur through the story, so we will report three length-normalised name quantities from the same universal-dependencies parse. They are the proportion of tokens tagged as proper nouns, the number of distinct proper-noun types per hundred words and the mean repetitions per type. Each norm-based mean is reported with and without the exclusion, together with the two versions' correlation within each language. Surprisal is computed with the names present in context and excluded only from the mean.

Measures

The processing outcome, serving Hypotheses 1 to 4 and 6, is the natural logarithm of elapsed reading time divided by the story's word count, standardised within reading language. Hypothesis 5 has a separate outcome, the child's optional post-reading rating of the story as easy, just right or difficult, treated as ordered and modelled on a cumulative-logit scale. A word, in the processing outcome's denominator and in every per-word quantity, is a whitespace-delimited token of the text as served. In nine of the ten languages, the token is an ordinary word. In Korean, it is an eojeol, a content word with its postpositional particles or verbal endings, which is the unit of the Korean dependency treebanks and of both parsers in Supplementary Table 3. A Korean text therefore contains fewer tokens than a translation of it would, so a Korean rate in words per minute is not comparable in content with the other nine. For the same reason, the plausibility bounds under Sampling plan and the fixed read-aloud bound under Confirmatory models are less strict for Korean at the fast end. Associations in Korean are interpreted through Korean's own language-varying slopes throughout.

The per-word scale also ties the outcome to word length. Mean orthographic length anchors LexicalDifficulty and helps determine how many words a text contains, since the same content in longer words needs fewer of them. As a result, time per word rises with word length even where time per character is unchanged. The per-character refit under Sensitivity analyses removes that dependence without altering the primary scale. Where the extract includes the platform's own word count, we will report its agreement with ours before any outcome model is fitted.

The outcome's intraclass correlations at the Text and the Account level will be reported from the tier-1 variance components. They give the reliability with which a text and an account are characterised by their readings.

The confirmatory predictors are the three component scores defined under Outcome-blind dimension reduction. Text length in words, narrativity, mean valence and mean arousal are pre-specified secondary text predictors and enter every model as covariates. Valence and arousal effects on word-recognition latency are larger among low-frequency words (Kuperman et al., 2014), so the affective means overlap LexicalDifficulty and enter only as adjustments. Omitted, they would leave part of that component's association unadjusted. No interaction between an affective mean and a component is estimated, and the variance inflation factors under Decision rules will show the overlap. Every affective value used here descends from adult ratings, a limitation recorded in Supplementary Section S3. Accordingly, these covariates are interpreted as properties of the text on an adult referent, not as what a child felt. The variability of valence and of arousal, and the additive and temporal connective densities, will be computed for description only. They enter no confirmatory model and are excluded from the component solution. The generated-or-classic indicator enters as a covariate, and the substantive comparison between generated and classic texts is a contingent hypothesis, as question (j) in Table 2 explains. Because predictors are standardised within language, a pooled coefficient averages slopes expressed in each language's own standard-deviation units. We will report the raw-scale standard deviation of each component per language so that coefficients can be converted to raw units, and the language-varying slopes summarise the heterogeneity this induces.

The age band is the platform's record of the child's age group, reduced to the two bands this study covers. An account whose recorded band changes within the window contributes each session under the band recorded for it, and the number of such accounts will be reported. The reading level is the

level the family has set for the session, which determines which version of a text is served. Narration is the session-level narration flag, recording that audio accompanied the reading.

Accumulated in-app reading, which serves Hypothesis 6, is the cumulative count of words in completed distinct stories the account read before the current session. Entering it as $\log(1 + \text{accumulated words})$ keeps it defined at the first eligible session, where the count is zero. The transformed variable is standardised within language, as every continuous predictor is, and only then decomposed into the account's mean and the current session's deviation from it. The registered test, the coefficient of the deviation term, measures the change in the standardised outcome per one within-language standard deviation of that deviation. Session order, the days since the account's previous eligible session and the days since its first eligible session enter the tier-2 model with the two terms.

The data are crossed: each reading belongs to one account, but a text is met by many accounts and exists in several languages and at several reading levels. An account may read in more than one language. We will treat Account, Text, Language, Country and App version as crossed grouping factors and report the counts supporting each before fitting any model. The grouping-factor rule admits a factor other than Text and Account when it has at least five levels with at least 20 eligible readings each. Otherwise, the factor is dropped, and the omission will be reported. Reading level, with only three levels, enters as a fixed factor. If at least 90 per cent of texts are read in only one eligible session, a Text varying intercept is not estimable and is replaced by varying intercepts for the repeated attribute combination that generated the text. These thresholds are registered conventions, and any consequence of them will be reported.

Randomisation and blinding

Nothing in this study is randomised, and no allocation, assignment or condition is under anyone's control.

Data collection and analysis will not be performed blind to the conditions of the study. The platform collects the data without researcher involvement. Before in-principle acceptance, however, we are blind to the outcome in the extract, which the data controller holds as described under Data provenance, sequestration and outcome blindness. After transfer, analysis will not be blind to the values of the predictors or grouping variables, since the models require them. We will write, run and debug the outcome-model code against data simulated from the design specification deposited with this protocol, so that every script is complete and tested before we receive the real extract.

Sampling plan

The design is observational, and the sample is a census. We will analyse every eligible record in one frozen extract, whose closing date is fixed before any outcome model is fitted, with no sequential testing or optional stopping.

Before the other sampling rules apply, we will exclude sessions the platform flags as incomplete or crashed, duplicated event records, accounts the platform flags as duplicates and stories without a positive word count.

Two nested samples are drawn from the remaining records, tier 1 for the text hypotheses and tier 2 for the within-child hypothesis, because these hypotheses are identified at different levels. Tier 1 comprises every completed reading whose account is recorded in either age band (7 to 8, or 9

and over) and whose reading language the feature pipeline covers. The story text must have been supplied, and the reading time must fall inside the plausibility bounds below. Where an account read the same text more than once, only the first reading enters tier 1. Repeated readings are retained for the practice analysis under Sensitivity analyses. The unit of analysis is one reading of one text by one account.

Under the Account rule, Account receives a varying intercept if at least 10 per cent of tier-1 readings come from accounts contributing two or more readings. If fewer do, the intercept cannot be separated from residual variance, so it is dropped and child variance stays in the residual. The sample's reduced effective independence is then stated with every tier-1 interval. The threshold is a registered convention, applied to the provider-computed counts before anything is fitted.

Tier 2 comprises the tier-1 accounts contributing at least three eligible readings of distinct texts. Three is the smallest number that leaves a degree of freedom once a within-child slope and an intercept are estimated. How many accounts qualify is not yet known.

The plausibility bounds on reading time per word are set by what a child could physically be doing, not by anything observed in these data. We will exclude a reading faster than 400 words per minute (0.15 seconds per word) as too fast to be reading. A reading slower than 10 words per minute (six seconds per word) will be excluded as a story left open. The fast bound is the top of the adult range and follows published precedent. Silent reading in educated adults typically runs between 200 and 400 words per minute (Rayner et al., 2016), so a child of 7 to 11 sustaining more than 400 is not plausibly reading every word. A meta-analysis of Spanish reading rates applied the same cut, treating rates above it as a sign that the reader had skipped part of the text. It removed 313 cases, all of them silent-reading rates, from tests that assessed 19,237 participants (Ripoll Salceda et al., 2020). The slow bound lies below the meta-analysis's slowest pooled mean, 30 words per minute for Spanish first graders reading silently, though above the normal-range floor proposed for that grade, 8 words per minute. We will report the number and proportion removed under each rule, by language, age band, reading level and narration state.

If the provider-computed counts show more than 1 per cent of sessions sharing one reading-time value, we will locate that value in the transferred extract before fitting any outcome model. We will treat it as a right-censoring point if it lies in the upper decile of the reading-time distribution. If so, the reading-time models will use a censored lognormal likelihood, with those sessions retained as censored.

No excluded datum is replaced. A participant-flow diagram will report every exclusion and the retained accounts, readings and texts, overall and by language.

Measurement, availability and missingness audit

Digital traces have an error taxonomy distinct from that of surveys (Sen et al., 2021), and four concerns here fall under the types it names. Three are trace measurement errors, in which the platform's record is an imperfect proxy for the construct. They concern the age-band and reading-level categories, instrument drift as the app changes version and a reading-time interval that is not fully documented, the subject of question (c) in Table 2. That drift is also a change in platform affordances, which the taxonomy treats as a separate type. The fourth concern is a trace selection error, since a session that times out is right-censored and never enters the extract.

Before fitting any outcome model, we will run and archive an audit of the frozen extract and its dictionary. The audit establishes field availability and units, the granularity of the age field, how often a text and a reading level recur, and the narration flag's behaviour within accounts and sessions. It checks duplicate and crash flags, changes across app versions, per-language feature coverage and every nesting and crossing count the models require. Missingness will be reported for every planned variable, overall and by language, age band, reading level, narration state and session order.

A feature structurally unavailable in a language is not imputed, so the language does not contribute it. A session missing an outcome is excluded from the model for that outcome alone, and a missing narration flag does not remove a session from a model without a narration term. No variable is selected, transformed or discarded because of its association with reading time or the rating.

The Hypothesis 5 rating is optional and was present in only 10 of the structural sample's 35 sessions, so we expect high missingness. If the rating is missing for more than 5 per cent of otherwise eligible sessions, we will model its availability from observed pre-response variables. In that case, we will report an inverse-probability-weighted analysis as a sensitivity check, showing its weight distributions and common support.

Design and precision analysis

Nature Human Behaviour's guidelines set 0.95 as the a priori power floor for frequentist plans. For inference by Bayes factors, they require either data collection until the Bayes factor reaches 10 in one direction or the other, or a maximum feasible sample at which an inconclusive result would still be an important message for the field. This study uses an estimation design on a census of existing records, so the second provision applies. With no further data to collect, the frozen extract is the maximum feasible sample. The design nonetheless takes its evidence threshold from the first provision. As Decision rules sets out, a non-negligible verdict requires an interval Bayes factor of at least 10 under the reference prior, the weakly informative prior registered under Priors. The assurance level at which a hypothesis counts as adequately powered is registered as 0.90, with every 0.95 requirement reported from the same archived curves (Supplementary Section S5). A hypothesis declared underpowered is still fitted and reported, so its interval bounds the effect even where it cannot decide the hypothesis. The assurance analysis is the fixed-sample counterpart of a Bayes factor design analysis (Schönbrodt & Wagenmakers, 2018).

The text coefficients are identified between distinct texts, so their precision depends on how many the extract contains, a number not yet known. To find how many are needed, we simulated data with pilotr from the deposited design specifications (Bernabeu, 2026b). The texts sweep varied the number of distinct texts at an average of twelve readings per text, the readings sweep varied readings per text at 250 texts and the tier-2 sweep varied the number of children. The simulated tier-1 model contained the three components, the age band, the two age-band interactions and a text intercept. The narration model contained the lexical and continuity components, the narration flag, their interactions with it and a text intercept. The tier-2 model had the three components, a single accumulated-reading trend and child and text intercepts, without the age band or its interactions. In the generated data, the text intercept contributed 9 per cent of the variance, the child intercept 20 per cent and the residual the remainder.

Data were simulated on the within-language standardised scale, so a simulated coefficient is directly comparable with the region of practical equivalence. The region's half-width, 0.05 standard devia-

tions, is a registered convention and the smallest effect this protocol treats as worth reporting. The simulated effect of 0.10 standard deviations, a second registered convention, is twice the half-width, because an effect at the boundary cannot be declared outside the region. A null condition with every coefficient set to zero established how reliably the design can declare practical equivalence.

The simulated models are simpler than the registered ones, and the simplifications bias the requirements in different directions. They omit the narration terms of the tier-1 model, the reading-level adjustment, the covariates, the language, country and app-version intercepts, the language-varying slopes, the distributional sub-model and the review-derived priors. The simulation generated child variance but analysed a dataset in which each account contributes one reading, so this variance entered the fitted model as inflated residual variance. In that one respect, the requirements below for the reading-time models are upper bounds. In the ordinal model for Hypothesis 5, the unmodelled variance also attenuates every coefficient towards zero, so the model's projection in Supplementary Section S5 is computed from the attenuated estimate and is not an upper bound. Adjusting for reading level, which moves all three feature families together, will remove some of the components' variance and raise every requirement, whereas the language-varying slopes and the priors will lower them. The prevalence-weighted main effects have the precision of the pooled slope and no design effect, but each interaction incurs the design effect for imbalance applied by the first two adjustments below. When the design analysis is re-run at the observed inputs, we will refit one grid point of each sweep with the registered brms specification, compute both registered criteria from those posteriors and report the difference.

The simulation scored its verdicts by criteria that lie between the registered posterior and Bayes-factor criteria. It counted a verdict as determinate when the 95 per cent interval lay wholly outside the region, and as equivalence when it lay wholly inside. An interval wholly outside the region leaves at most 2.5 per cent of a symmetric posterior inside it, whereas the posterior criterion allows up to 5 per cent. An interval wholly inside the region guarantees at least 95 per cent of the posterior inside it, which is the registered equivalence criterion.

For a non-negligible verdict, the Bayes-factor criterion is stricter than either the posterior criterion or the simulation's, so it sets the requirement. Under the reference prior, the interval Bayes factor against the region reaches 10 when the posterior mass inside it falls to 0.8 per cent for a main effect and 1.7 per cent for an interaction. For a normal posterior, those masses place the estimate 2.42 and 2.13 standard errors beyond the region's edge, against the interval's 1.96. The requirements under this criterion were computed from the same archived curves by that normal approximation, using the deposited script `design_analysis/solve_at_bf10.R` (checked against the simulation in Supplementary Section S5). The re-run at the observed inputs will compute this criterion by simulation from the brms refits and supersede these approximations.

Pairing the two criteria combines Bayesian estimation with Bayes-factor testing. These approaches differ in whether the model privileges a theoretically useful value such as the null, and each serves a different goal (Rouder et al., 2018). With the same region and prior, though, the interval Bayes factor and the posterior odds of the region come from one posterior and differ only in whether the prior odds are divided out (Campbell & Gustafson, 2023). Calibrated to the same error rate, the interval Bayes factor, the posterior rule and the frequentist equivalence test make almost the same decisions (Campbell & Gustafson, 2021; Linde et al., 2023). The choice between the two criteria therefore turns on which verdict each should decide, and reporting both is the current recommendation for

inferences about null and non-null effects (Heck et al., 2023; Lakens et al., 2020). The posterior rule has an independent decision-theoretic justification (Kelter, 2025).

For the three text coefficients (Figure 3, panel a), an effect of 0.10 standard deviations reached a determinate verdict nine times in ten at 738 distinct texts for lexical difficulty (95 per cent interval 640 to 843). Grammatical load needed 660 (593 to 731) and referential continuity 634 (568 to 704). At 0.95, the three coefficients needed 926, 812 and 784. The age-band interactions needed about twice as many: 1,230 texts for grammatical load (1,107 to 1,359) and 1,431 for referential continuity (1,276 to 1,596) at 0.90. At 0.95, the grammatical-load interaction needed 1,510 texts, and the continuity interaction did not reach that target within the 1,600 texts simulated. The narration contrast needed 1,234 texts (1,113 to 1,361) before the design effect for imbalance is applied, and 1,506 at 0.95. That requirement applies to the lexical-by-narration interaction coefficient. It is a lower bound for the difference-of-proportions rule of Hypothesis 4, because that rule's draw-by-draw ratio has heavier tails than the coefficient it is built from. Declaring practical equivalence when the true effect was zero required a similar number of texts. A main effect was correctly declared equivalent in 87 to 90 per cent of replicates at 800 texts and in every one at 1,600, but the interactions reached 89 to 92 per cent only at 1,600. An inconclusive verdict is therefore predictable from the number of texts alone.

For the unadjusted trend, the within-child term needed 609 children in tier 2 (528 to 697), each reading five distinct texts, and 731 (621 to 851) at 0.95. The registered term is adjusted for session order and elapsed time, with which accumulated words are correlated within a child, so this figure understates the requirement. The tier-2 re-run below re-derives it with those covariates.

Under the Bayes-factor criterion, the requirements for a non-negligible verdict rise. The three main effects need about 930 distinct texts for lexical difficulty, 930 for grammatical load and 940 for referential continuity at 0.90, and 1,120, 1,120 and 1,130 at 0.95. The grammatical-load interaction needs about 1,470 texts and the continuity interaction 1,480 at 0.90, neither reaching 0.95 within the 1,600 texts simulated. The narration interaction needs about 1,470 texts at 0.90, before the design effect for imbalance, and likewise does not reach 0.95 within 1,600. Tier 2 needs about 810 children at 0.90 and 970 at 0.95.

In the readings sweep, raising readings per text from 2 to 60 (Figure 3, panel b) narrowed the interval on a main effect from 0.19 to 0.08 standard deviations. Even so, a main effect's probability of a determinate verdict rose no higher than 0.70. Over the same range, that probability for the age-band interactions climbed from under 0.11 to between 0.87 and 0.90. Sessions and texts are therefore not interchangeable. A text coefficient is bounded by how many different texts the extract contains, whereas an interaction between a text property and a reader attribute gains from many different children meeting the same text.

The threshold each hypothesis has to clear is recorded in Table 1 and fixed before any part of the extract is transferred. Seven registered adjustments of the design analysis each draw on the provider-computed counts, the outcome-blind audit of the frozen extract or the archived prior-derivation pipeline. The first two are analytic corrections for imbalance. The simulation balanced the narration contrast, but narration was on in 32 of the 35 sessions of the structural sample. An unbalanced binary contrast at prevalence p inflates the variance of its interaction coefficient by $1/(4p(1-p))$ relative to balance. The first adjustment multiplies the simulated Hypothesis 4 requirement by

that factor at the observed narration prevalence, about 3.0 at a prevalence of 0.91, and states the adjusted requirement before fitting. The second applies the same factor to the Hypothesis 2 and 3 requirements at the observed age-band prevalence, since the simulation balanced the bands as well. The third re-runs the rating simulation at the observed rating availability and the fourth at the observed proportions of the three rating categories, both taken from the extract. The values published for an earlier window of this platform (Ratner & Murphy, 2026) cover ages from three years below this study's younger band, so they are not used as inputs. The fifth re-runs the tier-2 sweep at the observed distribution of sessions per child. For that re-run alone, the deposited tier-2 specification will be extended with session order and days since the first session. Per-session word counts will be generated so that accumulated words correlate with session order as the counts show. The extended specification will be deposited before it is run. The thresholds these three re-runs yield replace the ones in Table 1 for the underpowered declarations of Hypotheses 5 and 6.

The sixth adjustment runs only if the prior-derivation pipeline (Supplementary Section S6) requires it. Before any outcome datum is accessed, that pipeline will record whether each component's pooled estimate on the model's scale exceeds 0.10 standard deviations. Where a pooled estimate lies between 0.05 and 0.10, the texts sweep will be re-simulated with the deposited scripts at that effect in place of the 0.10 input. The resulting threshold replaces the one in Table 1 for that component before fitting. If a pooled estimate lies at or below the region's half-width, no number of texts could yield a determinate verdict at that size. The 0.10 convention is then retained, and the discrepancy will be reported with the component's verdict. The seventh adjustment runs only if the compression diagnostic falls below its cut-off. It re-runs the texts sweep at the observed number of unnarrated readings per text, taken from the audit, and yields the threshold for the narration rule stated with Hypothesis 1. That threshold is stated before any Hypothesis 1 verdict is taken from an unnarrated slope.

Analysis plan

We will fit all confirmatory models as Bayesian hierarchical models with the brms package and its Stan backend (Bürkner, 2017), standardising continuous predictors within language after transformation. Where the accompanying systematic review supports it, weakly informative priors will be replaced by review-derived priors, as described under Priors.

Outcome-blind dimension reduction

The observed text features are correlated because the reading level a text is written for moves lexical, grammatical and discourse properties together. We will reduce them by principal component analysis, separately within the lexical, grammatical and discourse families, without treating them as effects of an unobserved latent cause (Crossley, 2025). The affective measures will be kept as separate secondary predictors, because valence and arousal are theoretically distinct and need not move together. Narrativity, though computed as a discourse feature, will likewise be excluded from the component solution.

Each distinct text will contribute one row however often it was read, so a popular text does not determine the solution. Within a family, a text enters the solution only if it has valid values on every retained feature. A text with fewer than five sentences, as segmented under Materials, enters no family's solution. The theory object records this minimum as the boundary condition for referential continuity. Since the tier-1 model includes all three components, a text below the

minimum contributes to no confirmatory model, so the thresholds in Table 1 apply to the distinct-text counts that remain. We will report, per language, the distribution of sentences and of words per text and the number of texts the minimum removes.

Counts will be transformed with $\log(1 + x)$ and bounded proportions with an empirical-logit transform where these reduce skew. A feature is dropped from a language if more than 20 per cent of its values are missing or its standard deviation after transformation is below 0.01. The text-coverage rule drops a feature from any language in which fewer than 80 per cent of texts receive a valid token-level match. A language contributes to a family's solution only if it has at least 100 distinct eligible texts and at least three valid features in that family, two of them anchors (defined below). Where two features correlate at $|r| > .90$, the retained one is chosen in this order: lower missingness, then broader language coverage, then higher extraction reliability on a manually checked sample, then the measure listed first under Materials. No outcome correlation enters any of these decisions.

To stop the largest language determining the solution, we will compute a correlation matrix within each eligible language, transform the correlations to Fisher's z and average them with equal weight per language. The average will then be transformed back for analysis (Revelle, 2023). Sampling adequacy requires individual Kaiser-Meyer-Olkin values above .50 and an overall value above .60. Parallel analysis (Horn, 1965) will set the number of components, retaining each component whose eigenvalue exceeds the 95th percentile of the corresponding eigenvalues from 10,000 random datasets. That percentile criterion, Glorfeld's modification of Horn's average (Glorfeld, 1995), is the more conservative of the two. The scree plot will be reported but will not override that rule. Where a family retains more than one component, the components will be rotated obliquely with promax, because linguistic properties co-occur and do not vary independently (Biber, 1988). Promax will use $\kappa = 4$, the default of the implementation used (Revelle, 2023). We will then apply the regression scoring coefficients from the frozen, language-balanced solution to each text's features, standardised within language.

The anchors of each confirmatory component are the features that identify it. For LexicalDifficulty, they are frequency, surprisal and word length. Surprisal is grouped with the lexical features because it is hardest to separate from word frequency at the word level (Oh & Schuler, 2025). Model surprisal also diverges most from human reading times on the rarest words (Oh et al., 2024). The correlation between surprisal and each grammatical feature will be reported per language next to the pattern matrix, but the family assignment fixed here does not depend on it. For GrammaticalLoad, the anchors are dependency distance, subordination, clause density, sentence length and noun-phrase depth. For ReferentialContinuity, they are adjacent-sentence similarity, referential overlap and the densities of causal and contrastive connectives. Age of acquisition enters the lexical family where a broad native norm set exists, but it is not an anchor, because the cross-linguistic set covers only 299 words per language. Only the causal and contrastive connective densities enter the discourse family, and both are anchors. The additive and temporal densities are descriptive only, as stated under Measures, because causal and contrastive connectives raised comprehension, temporal ones had no effect and additive ones lowered it (Kleijn et al., 2019). A count mixing all connective types therefore has no defined expected sign. Each component's sign is set to match the orientation stated in the Introduction.

Under the construction rule, each confirmatory component must load at least .30 in absolute value on at least two of its anchors. If no component in a family meets this rule, the hypotheses resting

on that family are reported as untestable. Where more than one component meets the rule, the one with the most anchors reaching the .30 threshold is confirmatory, and the others are reported descriptively. In the discourse family, a tie is broken in favour of the component on which the causal and contrastive densities load highest, because the cue-utilisation account that Hypothesis 3 adjudicates is stated for connectives. The tie-break, a registered convention, implies no claim that connectives are the more predictive cue, since in the child sample closest to this one, deep and referential cohesion both explained unique variance in comprehension (Pickren et al., 2022). If the component so chosen loads below .30 on both connective densities, Hypothesis 3 is still tested on it, under the narrower interpretation given in Table 1.

Concreteness needs a separate rule, because it has a usable rated set in nine of the ten languages but is not an anchor for LexicalDifficulty. It is dropped from Korean under the missingness rule above and from any other language where it fails the token-coverage rule under Materials. In the language-averaged matrix, its correlations draw only on the languages that retain it. The lexical solution will be reported with and without it, alongside Tucker's coefficient of congruence between the two.

The component solution will be reported in full: the pattern matrix, the component correlations, the variance explained, the language-specific matrices, the loading congruence for the five largest languages and the raw-scale standard deviation of mean dependency distance per language. Language-specific and sparse solutions are sensitivity analyses and will never replace the primary solution after an outcome is seen.

Confirmatory models

In tier 1, we will estimate a model of within-language standardised log reading time per word. Its fixed effects are the three components, the age band, the narration flag, each component's interactions with the age band and with narration, the reading level, log text length, narrativity, valence, arousal and the generated-or-classic indicator. The age band and narration are coded as deviations from their observed prevalence. The model has varying intercepts for Text, Country and App version, and varying intercepts and uncorrelated component slopes for Language. Account receives a varying intercept where the Account rule under Sampling plan is met. Hypotheses 1 to 4 will be evaluated in this model. Because the model includes every component-by-age-band and component-by-narration interaction, each registered interaction is adjusted for the others, and each main effect is the prevalence-weighted average slope defined in the Introduction. Three of those terms have no registered test: LexicalDifficulty × AgeBand, GrammaticalLoad × Narration and the components' remaining narration terms outside the Hypothesis 4 contrast. They will be reported with posterior medians and intervals but no decision rule, so that no term is reported selectively. None will be taken as evidence about whether the age band stands in for a bundle of developing capacities. A component's slope in a given state is recovered draw by draw as the main effect plus the interaction multiplied by that state's deviation score.

The Hypothesis 4 contrast is computed from these state-specific slopes. For each posterior draw, the proportional attenuation of a component is one minus its narrated slope divided by its unnarrated slope, so a positive value means a smaller coefficient under narration. The contrast is computed only under the Hypothesis 4 precondition, which requires fewer than 5 per cent of the draws of each derived unnarrated slope to lie inside the region. Draws in which either unnarrated slope lies inside the region will be excluded from the ratio, and their fraction reported. As a sensitivity analysis,

the contrast will be repeated with a fixed denominator, each narration interaction divided by the posterior median of its unnarrated slope, and reported with the draw-by-draw version.

The first of three further rules for Hypothesis 4 concerns accounts observed in both narration states. If they cover at least as many distinct texts as the adjusted requirement, the within-account contrast restricted to them becomes the primary estimate, and the full-sample contrast is reported with it. If they cover fewer, the within-account contrast remains a sensitivity analysis, and selection into narration state is acknowledged wherever the full-sample estimate is reported.

Second, the tier-1 model contains a distributional sub-model in which the narration flag also predicts the residual standard deviation. The ratio of the narrated to the unnarrated residual standard deviation is the compression diagnostic. A ratio below 0.5 is the registered signature of the audio setting the pace. When the diagnostic falls below that threshold, the full-sample verdict is withheld and Hypothesis 4 is evaluated in the subset described next. For each language, we will report the share of narrated sessions whose rate lies within 10 per cent of that language's median narrated rate, because the ratio can miss a minority of audio-paced sessions. This share is descriptive and enters no decision rule.

Third, the contrast will be repeated in the narrated sessions the audio cannot have paced. For each language, the narration rate will be estimated as the median words per minute of its narrated sessions, and a narrated session counts as unpaced when its rate exceeds 1.5 times that median. Where a language has fewer than 100 narrated sessions, a fixed bound applies instead. The bound is that language's adult read-aloud mean in the International Reading Speed Texts or, for a language they do not cover, the slowest of their 17 means, 138 words per minute for Arabic (Trauzettel-Klosinski & Dietz, 2012). For Korean, the Arabic mean is applied to eojeol counts and so becomes the least strict of the fixed bounds. We will report the size of the unpaced Korean subset this bound defines. The premise is that narration produced for children is paced no faster than adult oral reading in the same language. If the selectivity persists in the unpaced subset, pacing does not explain it. Hypothesis 4 is reported as underpowered from the subset if the subset contains fewer distinct texts than the adjusted requirement. Should the data controller supply the per-text audio duration, or disclose a fixed rate at which narration plays, that quantity replaces these rules for defining an unpaced session. We will report the substitution as a protocol deviation.

The Hypothesis 5 model will have the same fixed and grouping structure and a cumulative-logit likelihood over the three rating categories, with category thresholds that vary by language. Cross-language differences in use of the middle category are then not mistaken for differences in text difficulty. Whether the Account intercept is estimable in this model depends on how many accounts contribute more than one rated reading, which the tier-1 count does not establish. We will report that number and, where the intercept is not estimable, acknowledge child-level scale use as a limit on the rating. Band-specific odds ratios, derived from the component-by-age-band terms, will be reported descriptively alongside the pooled ones. A text rated by fewer than five distinct children does not contribute, and the primary model is complete-case, with the inverse-probability sensitivity analysis registered under the audit.

For tier 2, we will refit the same specification with a varying intercept for Account and decompose accumulated reading into the account's mean and the current session's deviation from it (Curran & Bauer, 2011). Session order, days since the previous session and days since the account's first

eligible session enter as covariates. The deviation term tests Hypothesis 6.

Posterior predictive checks will assess calibration and residual structure by language, age band, reading level and narration state in every model.

Priors

Priors for each confirmatory main effect, and for the secondary text predictors for which the review has at least two studies, will be derived from the systematic review by a deterministic pipeline. We froze its rules before the review's data extraction began. We will run it and archive its output before any outcome datum is accessed. The review's estimates for the corresponding predictor family and outcome class will be sign-harmonised so that facilitation means faster reading, converted to a common standardised scale, and pooled by random-effects meta-analysis with one estimate per study per relation. The prior is Student's $t(3, \hat{\mu}, \sigma)$, where $\hat{\mu}$ is the pooled estimate translated to the model's scale and σ is twice the combined pooling and between-study uncertainty, with a floor of 0.10. The doubling and the floor guard against the gap between a laboratory comprehension test and a passively timed reading in a game.

The same review motivates the hypotheses, so the evidence that generates a prediction also centres its prior. Two features of the decision rule keep that from being circular. First, the verdict's posterior criterion is evaluated on the derived-prior fit, but its evidence criterion, the interval Bayes factor defined under Decision rules, is computed from the weakly informative refit described next. A prior centred in the predicted direction therefore does not supply the evidence that confirms the prediction. Second, a hypothesis whose verdict changes under the weakly informative refit is reported as sensitive to the prior, and its result is not counted as conclusive. We will report prior-data conflict diagnostics for every review-derived prior.

Coefficients without adequate review evidence, namely every interaction term and any family supported by fewer than two studies, keep weakly informative priors: $t(3, 0, 0.5)$ for standardised main effects and $t(3, 0, 0.25)$ for interactions. Group-level and residual standard deviations take half-$t(3, 0, 1)$ priors, and any estimated correlation matrix takes LKJ($\eta = 2$). Intercept priors will be calibrated by prior-predictive simulation to the outcome scale, without reference to any predictor-outcome association. The derivation rules are summarised in Supplementary Section S6. The full prior table and its forest plots will be archived before the extract is transferred and will accompany the Stage 2 report. We will refit every model with weakly informative priors throughout, and with the derived scales halved and doubled, and report conclusions under each.

Decision rules and diagnostics

For every confirmatory coefficient, we will report the posterior median, the 95 per cent highest-density interval, the probability of the predicted direction, the posterior proportion inside the region of practical equivalence and a model-implied contrast on an interpretable scale. The region is $|\beta| < 0.05$ standard deviations for a one-standard-deviation change in a predictor of reading time, and a half-width of log(1.05) on the log-odds scale for the rating model. More than 95 per cent of the posterior inside the region counts as practical equivalence. A practically non-negligible association is labelled in the predicted or opposite direction by the sign of the posterior median. It requires less than 5 per cent of the posterior inside the region and an interval Bayes factor of at least 10 for the region on that side against the region inside. Any other result is inconclusive.

The Bayes factor is the ratio of the posterior odds of the two regions to their prior odds (Hoijtink et

al., 2019; Klugkist et al., 2005; Morey & Rouder, 2011). The posterior odds come from the draws of the weakly informative refit, whose Student-*t* prior is the reference prior for evidence. The prior odds come from that prior's mass in the region, known in closed form. No marginal likelihood is needed, so the Bayes factor avoids the fragility that marginal-likelihood Bayes factors show in mixed models (Schad & Modrák, 2026; van Doorn et al., 2023). For the two-sided interactions of Hypotheses 2 and 3, it is computed for each side against the region, and the verdict states the side the evidence favours. Every Bayes factor is reported alongside its verdict, with its values under the derived priors and their halved and doubled scales, from the refits registered under Priors.

Which of the two criteria is stricter depends on the kind of verdict. Under the reference prior, 95 per cent of the posterior outside the region corresponds to a Bayes factor of only 1.5 for a main effect and 3.2 for an interaction. These values are low because the prior already places most of its mass outside the region, so the Bayes factor decides non-negligibility. For equivalence, 95 per cent inside corresponds to a Bayes factor above 100 for the region. A Bayes factor of 10 alone, by contrast, would be reached with less than half of the posterior inside it, so the posterior mass decides equivalence (Kruschke, 2018). Taking the stricter criterion for each kind of verdict keeps the journal's evidence threshold for Bayes factors without letting a diffuse prior produce an equivalence verdict.

The same two criteria apply to the Hypothesis 5 odds ratios on the log-odds scale. For the Hypothesis 5 ordering clause, a posterior probability of at least 0.90 that the three coefficients rank as they do for reading time counts as agreement on order, and anything lower as order not established. The Hypothesis 4 contrast, a derived quantity, is judged by three posterior criteria. They are at least 95 per cent of the posterior of the difference of attenuations above 0.20, more than 95 per cent inside the region and more than 95 per cent below -0.05. Table 1 gives the interpretation of each outcome, and a posterior meeting none of the criteria is inconclusive. We will report the interval Bayes factors of the contrast's two constituent narration interactions. The fixed-denominator sensitivity contrast is evaluated against the same thresholds.

Each hypothesis is a single registered test with its own decision rule, and no claim aggregates evidence across hypotheses, so no correction for multiple comparisons will be applied. The one composite claim, the combined Hypothesis 1 verdict, follows an intersection-union rule. It is recorded only when all three coefficients are individually non-negligible in the predicted direction, so its error rate cannot exceed that of any single component. Hierarchical shrinkage and the null condition of the design analysis, which reports how often a zero effect is wrongly declared non-negligible, calibrate the per-decision error rates behind Table 1.

Six registered contingencies depend on quantities computed after fitting, each a conditional rule stated where it applies. Two can leave a hypothesis without a verdict. Failure of the Hypothesis 4 precondition leaves Hypothesis 4 not evaluable, and failure of the positive control on recorded reading time leaves Hypotheses 1 to 4 and 6 not evaluable. Of the other four, the compression diagnostic, when it indicates compression, withholds the full-sample Hypothesis 4 verdict in favour of the unpaced subset. Under the narration rule stated with Hypothesis 1, the same diagnostic can move a component's Hypothesis 1 verdict to its derived unnarrated slope. Draws with an unnarrated slope inside the region are excluded from the Hypothesis 4 ratio. The third outcome-neutral check (next section) makes the within-level estimate primary for a component whose within-level coefficient falls below half its pooled magnitude. The prior-sensitivity rule above applies when a

verdict changes under the weakly informative refit. Two further rules respond to the fitting itself and not to any estimated association. Models failing the sampling diagnostics are simplified in the order given below, and a failed permutation check withholds the confirmatory results until the source of the structure is found and reported.

We will run four chains of 4,000 iterations, half of them warm-up. Acceptable sampling requires $\widehat{R}$ below 1.01, bulk and tail effective sample sizes above 1,000 for every confirmatory parameter, no divergent transitions, satisfactory energy diagnostics and no unresolved maximum-treedepth warnings. Models failing these criteria will be refitted with a higher target acceptance rate, greater tree depth and more iterations. If simplification is still needed, language-level slopes will be removed in the fixed order lexical, then discourse, then grammatical, and every estimable varying intercept will be retained. That order follows the expectations set out in the Introduction. We will report every simplification and its diagnostic consequence. Variance inflation factors will be reported for the fixed-effect design matrix, but a high value will not prompt an outcome-dependent deletion. The joint model remains primary, and single-family models will be reported as a collinearity sensitivity analysis.

For Hypotheses 2, 3 and 4, we will compare the model with one omitting the relevant interaction, using Pareto-smoothed importance-sampling leave-one-out cross-validation. We will report the expected log pointwise predictive density difference, its standard error and the Pareto-k diagnostics, with exact refits where importance sampling is unreliable. Predictive comparison supplements the coefficient-level summaries and decides no verdict.

Outcome-neutral quality checks

Six checks are registered, each with a pass criterion and a consequence of failure. All but the fifth run before the confirmatory models are interpreted. The fifth, convergent validity, is the ordering clause of Hypothesis 5 and is evaluated with that hypothesis, which states the check's outcome if the hypothesis is underpowered. The first, second, fourth and sixth are decided without reference to any confirmatory coefficient, whereas the third is a robustness condition on the fitted text coefficients.

For the first check, a permutation null, we will reassign texts at random to readings within language and reading level, and refit the tier-1 model 200 times. The refits will use the maximum-likelihood analogue of that model, with the same fixed and estimable grouping structure, and the region will be applied to the 95 per cent Wald interval. We choose this approximation because 200 full Bayesian refits are not feasible and the check concerns structure introduced by the pipeline. No text coefficient may be declared practically non-negligible in more than 5 per cent of those refits. An inconclusive verdict on permuted data is expected at moderate text counts and is not a failure. Failure would mean the pipeline has introduced structure that does not come from the pairing of a text with a reading. In that case, we will withhold the confirmatory results until the source of that structure is found and reported.

The second check, a positive control on recorded reading time, requires total elapsed time to rise with the story's word count, with more than 99 per cent of the posterior above zero. Failure means recorded reading time does not index reading, and Hypotheses 1 to 4 and 6 are reported as not evaluable on this extract.

The third check asks whether the association is an artefact of platform assignment. We will re-estimate the three text coefficients separately within each reading level, and each must retain at

least half its pooled magnitude. Where a coefficient falls short, the within-level estimate becomes primary for that component. The pooled estimate is then reported alongside it, and the assignment confound is named wherever the coefficient appears.

The fourth check tests three independence implications of the theory object. First, conditional on the reading level and the generated-or-classic indicator, the three components should be close to independent across texts. Second, narration should be unrelated to accumulated reading. Third, the difficulty rating should carry no further information about reading level once reading time is conditioned on. We will report each as an estimate with an interval. An implication fails when the 95 per cent interval of its conditional association lies entirely beyond 0.10 in standardised units. Such a failure does not invalidate the confirmatory tests and is reported as showing a path the causal graph omits. For the third implication, the omitted path would run directly to the reported difficulty from the reader's capacity, which the reading level partly indexes, or from the generation route. That path would be stated with the Hypothesis 5 odds ratios. Supplementary Section S1 gives the literature's reasons to expect this implication to fail, but the same criterion applies.

The sixth check validates the feature pipeline by hand. In a stratified random sample of 50 texts per language, we will inspect tokenisation, sentence splitting and dependency attachment, and report agreement per language before any feature is linked to outcome data. A language whose labelled-attachment agreement with the hand-checked parse falls below .85 contributes to descriptive reporting but not to the grammatical component.

Sensitivity analyses

Eight families of sensitivity analysis are registered, all of them descriptive. Each bounds a primary estimate without replacing it, and none changes which hypothesis is tested or how its verdict is reached.

In the first family, we will refit the reading-time models with no plausibility bounds, with winsorisation at the 1st and 99th and at the 0.5th and 99.5th percentiles within language, and with posterior-predictive residual trimming. Where the censoring rule applies, we will report the censored and uncensored fits side by side. The tier-1 model will be refitted with the outcome per character, and its three component coefficients reported with the primary estimates.

The second family redefines eligibility, moving the tier-2 threshold from three distinct texts to two and to five and, as a practice analysis, adding repeated readings with a repetition indicator. With the sentence floor raised to eight, we will refit the tier-1 model and report its three component coefficients and distinct-text count with the primary estimates. Grouping by Text will be compared with grouping on the attribute combination that generated the text. The classic stories are likely to be public-domain texts, which the surprisal model's training corpus could contain (Supplementary Section S3). We will therefore report the mean per-word surprisal of classic and of generated texts per language before any outcome data are linked. For the same reason, we will refit the tier-1 model on generated texts alone, reporting its three component coefficients with the primary estimates.

The third family examines variation between languages, beginning with Hypothesis 6. Where Language, within tier 2, meets the grouping-factor rule under Measures, we will refit the tier-2 model with an uncorrelated language-varying slope on the accumulated-reading deviation and report the per-language slopes. If that refit fails the sampling diagnostics under Decision rules, it will be dropped and the drop reported. We will restrict analyses to the five largest languages and fit

language-specific models and component solutions, reporting their loading congruence. From those fits, we will report the two age-band interactions per language next to the pooled coefficients, arranged by orthographic depth where a published classification places the language. A further tier-1 refit will let both interactions vary by Language. The design analysis computed no requirement for the language-specific interactions or the language-varying refit, and both will give wide per-language posteriors. Neither supplies a second test of Hypothesis 2 or 3. The simplification order does not remove the language-varying interaction slopes, because they lie outside the tier-1 specification.

The fourth family asks how much the estimates depend on who is reading in which language. Before the transfer, we will deposit a lookup table marking whether each reading language is an official or majority language of each country, as a coarse indicator of the match between the two. We will report the cross-tabulation of reading language by account country and the number of accounts reading in more than one language. We will then refit the tier-1 model on sessions whose reading language is a language of the account's country. Its three component coefficients will be reported with the full-sample estimates, and its distinct-text count with the threshold in Table 1. The indicator enters no confirmatory model. Because it misclassifies in both directions, the refit bounds the estimates' sensitivity to sample composition and identifies no individual reader.

The fifth family rescores the text properties, replacing the component scores with sparse-component scores and with pre-specified equal-weight composites of the anchor features. For Hypothesis 3, we will report the age-band interaction of each ReferentialContinuity anchor from a model in which that anchor replaces the component, to check whether the three devices agree in sign.

The sixth family varies the age contrast in Hypotheses 2 and 3. Where the audit finds the age field finer than two bands, we will re-estimate both age-band interactions with the finer age entered in two ways and report each coding's estimates with the two-band contrast. One coding treats the finer age as an ordinal monotonic predictor (Bürkner & Charpentier, 2020), the other as an unordered factor contrasting each finer band with the band for ages 7 to 8. Pages per hundred words, computed from the recorded page count, will enter a tier-1 refit along with its age-band interaction. The Hypothesis 2 and 3 interactions from that fit will be reported with the primary ones, to gauge how much of the age difference is time away from the text.

The seventh family adds analyses for Hypotheses 4 and 5. For narration, we will report the within-account contrast, where it is not already primary, and the fixed-denominator contrast registered under Confirmatory models. We will also report the Hypothesis 4 contrast within each age band and the duration-adjusted contrast from the refit that includes the predicted spoken duration per word and its interaction with narration. For the rating, we will compare the model under adjacent-category and cumulative-logit links, stratify it by language and set the inverse-probability analysis for rating non-response against the complete-case estimates.

Finally, the eighth family applies alternative regions of practical equivalence: 0.03 and 0.10 standard deviations for reading time, and half-widths of log(1.03) and log(1.10) on the log-odds scale for the rating. Interval Bayes factors will be recomputed for each region.

To ease comparison with earlier studies, we will report maximum-likelihood mixed models with the same fixed and estimable grouping structure in a supplementary table. They are descriptive and will decide no Bayesian verdict.

Questions outstanding at submission

We put thirteen questions about the extract to the data controller on 7 August 2026, and they remain open at submission. Five of them correspond to auxiliary assumptions in the theory object. Table 2 lists the questions under their original letters, (a) to (m), and gives each a decision rule that depends only on its answer. The table opens with one further unknown, the number of distinct texts.

Table 2

Questions outstanding at submission and the decision rule for each answer

Open question	Bearing on the design	Decision rule for each answer
How many distinct texts have eligible readings. Materials gives the data controller's estimate and the working assumption.	Every text coefficient, since all are identified between texts.	The number will be established before modelling from the provider-computed counts and the answer to question (d), and placed on the precision curve in Figure 3. Below the threshold in Table 1, a hypothesis is declared underpowered before its model is fitted. Where each session has its own text, the Text varying intercept is replaced by varying intercepts for the repeated attribute combination, under the rule for texts read only once, given under Measures. Precision then depends on the number of attribute combinations. We will report whether sessions share texts.
Whether the narration flag records that audio played or only that it was available, and whether it can change within a session. The rest of the question covers whether the voice is recorded or synthesised, how many voices serve a language and whether words or lines are highlighted in time with the voice. It then covers whether a page advances automatically when playback ends, whether playback can be paused or repeated, the narration rate and the per-text audio duration (question a).	Hypothesis 4, through the auxiliary assumption that narration played.	If the flag records playback, Hypothesis 4 is tested as written. If it records availability only, the variable is relabelled as availability, and the contrast becomes an intention-to-treat comparison, whose attenuation is diluted towards the null by non-compliance. That dilution is stated wherever the coefficient is reported. Sessions in which the flag changes are excluded and counted. If pages advance automatically when playback ends, narrated sessions are the case the compression diagnostic under Confirmatory models is designed to detect. The diagnostic's registered consequence then applies: the full-sample contrast is not interpreted, and Hypothesis 4 is evaluated in the unpaced subset. If words or lines are highlighted in time with the voice, the highlighting is reported as a boundary condition wherever a narration coefficient appears, and no coefficient changes. If a fixed narration rate or the per-text audio duration is disclosed, the substitution registered in the third Hypothesis 4 rule applies.
Whether one story identifier corresponds to one text, and whether a text differs across reading languages by translation or independent generation (question d).	The identity of the stimulus, through the auxiliary assumption that the keyed text is the text the child read, and with it every model's grouping structure.	If one identifier is one text, texts are counted by identifier crossed with language and reading level. If a text is translated across languages, a varying intercept for the underlying story is added, crossed with Language. If each language is generated independently, texts in different languages are distinct and no such term is added.
Whether a stable identifier exists for each session. The structural sample has none (question e).	The matching of sessions to texts and the check of the transferred file against its manifest.	If one exists, it is included in the extract and used for matching. If none exists, sessions are matched to texts on story identifier, reading language, reading level and app version, as Materials states, and the transferred file is checked against its manifest by row count and checksum. Neither answer changes a model.

Open question	Bearing on the design	Decision rule for each answer
Whether the reading-time interval includes embedded activities, backgrounded time or idle time (question c).	The outcome, through the auxiliary assumption that elapsed time indexes effort.	If the interval excludes all three, the outcome is used as defined. If it includes embedded activities, texts that offer them are identified from the story attributes, an indicator for those texts enters every model, and the primary analysis is repeated within texts that offer none. If it includes idle or backgrounded time, the 10-words-per-minute plausibility bound and the censoring rule become more important, and the limitation is stated in the abstract.
Whether an account persists across reinstalls and whether siblings can share one (question f).	Tier 2 and Hypothesis 6, through the auxiliary assumption that an account is a child.	If an account can be shared, the within-child interpretation of Hypothesis 6 is withdrawn in advance, and the coefficient is reported as a within-account association, with the ambiguity stated in the hypothesis.
Whether the reading level can change within an account over time, and whether it changes the text served or only the page count (question g).	The platform-assignment confound and the third outcome-neutral check.	If a change of level alters only pagination, texts are comparable across levels, and the pooled estimate is primary, with the within-level estimate as a check. If it alters the text, the within-level estimate becomes primary and the pooled one is reported alongside it. A within-account contrast on a level change is not registered, so any such analysis would be reported at Stage 2 under Exploratory analyses.
The source of the age band and how much of it is missing across the whole window. It was blank for 10 of 35 sessions in the structural sample (question h).	Hypotheses 2 and 3.	Missingness will be reported by language, reading level and app version. Above 20 per cent, availability is modelled from observed pre-response variables, and an inverse-probability-weighted fit is reported and interpreted together with the complete-case fit.
Which reading languages the window covers. The app is marketed in 19 languages and the structural sample showed ten (question i).	Coverage of the feature pipeline and the language grouping factor.	The observed set will be reported. A language lacking the required resources contributes to the participant-flow diagram and the descriptive tables but not to any feature-dependent model, and the excluded volume is quantified.
Whether the classic stories are fixed, human-authored texts identical for all users within a language (question j).	The comparison between machine-generated and human-authored material, set out below as contingent hypothesis C4.	The condition was not confirmed before submission, so C4 is not activated, and the generated-or-classic indicator remains a covariate.
Whether unfinished sessions are retained anywhere, and what the completion rate is (question k).	Generalisability, since every row in the extract is a completed reading.	The results are stated to apply only to completed readings. Where an aggregate count of unfinished sessions by language is available, the completion rate is reported and the restriction quantified.

Open question	Bearing on the design	Decision rule for each answer
What fields the older data contain and what window they cover (question l).	Hypothesis 6, through the within-window assumption that accumulated reading acts on decoding automaticity over the observation window and not the reverse. The measure is truncated by a window opening on 1 January 2026.	If the older data contain the same fields, accumulated reading is computed over the longer window, and the analysis is repeated on both. Otherwise, the measure is left-truncated at the window's opening and is described throughout as reading accumulated within the observation window. The truncation makes a child who read in the app before the window opened look like a new reader.
Whether post-story responses include comprehension items or only the difficulty rating (question m).	Contingent hypotheses C1 to C3.	The condition was not confirmed before submission, so C1 to C3 are not activated, and no comprehension claim is made among the confirmatory results. The rule for a later confirmation is given under Contingent hypotheses.
What the per-session activity counters measure. In 6 of the 14 structural-sample sessions with activity, they sum to more than the recorded reading time, in one case to nearly twice it (question b).	None, by decision.	These fields are not used as a predictor, a covariate or an exclusion criterion. The decision is recorded here so that any later use would be visible as a deviation.

Contingent hypotheses

Questions (j) and (m) bear on four further hypotheses. We specify them in full but outside the confirmatory set, because each depends on a fact about the extract that the data controller has not yet confirmed. Each has an activation rule based on that fact alone, established from the data controller's written answer or, for the requirement that accounts be observed in both narration states, from the provider-computed counts.

A contingent hypothesis is activated only if its condition is confirmed in writing by the data controller before submission, and the confirmation is reproduced in Supplementary Section S7. At submission, no condition has been confirmed, so none of the four is activated, as that section records. If a condition is confirmed after submission, its hypothesis is reported at Stage 2 under Exploratory analyses and is not counted among the confirmatory tests. We retain the four hypotheses so that any such analysis is the one written before we had any access to the extract's outcome data. A hypothesis whose condition is never confirmed is not run.

C1: coherence and inferential comprehension. Its condition is that item-level comprehension responses exist and include a recoverable question type. In an item-level model of response correctness, with inference questions coded 1 and literal questions 0, we would predict a positive ReferentialContinuity × QuestionType interaction. The association of continuity with correctness would then be larger where the question requires integration. We found no study that tests a continuity by question-type interaction directly, which is one reason this hypothesis is contingent. The prediction therefore combines three results: an inference question is answered by integrating information across clauses (Freed & Cain, 2021), cohesion predicts children's comprehension overall (Pickren et al., 2022), and text properties bear differently on surface and inferential questions (Aparicio et al., 2022).

C2: narration and comprehension. Its condition is that item-level comprehension responses exist and that accounts are observed both with and without narration. We would predict that, within an account, narration is positively associated with response correctness after adjustment for text, age band, reading level, accumulated reading and language. The expected size is small (Clinton-Lisell, 2023; Wood et al., 2018), and the coefficient identifies an association, not an effect.

C3: reading pace and comprehension. It has the same condition as C2 and is restricted to sessions without narration. We would predict that an account's mean log reading time per word is negatively associated with response correctness, so that accounts with a faster habitual pace answer more accurately (Sortkær et al., 2021). The within-account deviation term in the same model separates this between-account association from slow reading in a difficult or interrupted session.

C4: machine-generated and human-authored text. Its condition is that the classic stories are confirmed to be fixed, human-authored texts, identical for all users within a language. We would predict that the three text components have coefficients of the same sign for reading pace in generated and classic texts. Each component's interaction with the generated-or-classic indicator is predicted to be practically equivalent to zero. The hypothesis tests whether the two kinds of text are exchangeable. The closest evidence is the adult eye-movement corpus on machine-generated text (Bolliger et al., 2025). A practically non-negligible interaction would mean that a text property relates differently to children's reading pace in generated and human-authored stories.

Timeline

The extract is built and held before submission, so the study can begin on the day in-principle acceptance is confirmed. Transfer is expected within four weeks of that confirmation. The audit of the frozen extract, the feature extraction with its hand-checked validation and the outcome-blind dimension reduction will follow within three months of transfer, and the confirmatory and sensitivity models within a further four months. The Stage 2 manuscript will be submitted within twelve months of in-principle acceptance. [The author to confirm or adjust these intervals.] Any extension would be negotiated with the handling editor.

Data availability

The data controller holds the session-level extract. It will be transferred to us only after the journal confirms in-principle acceptance, under a data-sharing agreement between the University of Oxford and the data controller. On acceptance of the Stage 2 manuscript, we will deposit the story texts, computed text features, component scores and fitted model objects in [the named public repository, with a digital object identifier]. The deposit will include the participant-flow counts, the laboratory log and every table and figure underlying the reported results. Session-level fields that the agreement classes as commercially sensitive may be restricted. For each such field, we will state the reason and deposit a derived form sufficient to reproduce every reported analysis. Researchers will be able to request the restricted fields through [the named data controller contact or the mechanism the agreement specifies].

The manifest and the provider-computed counts contain no record of any individual and will be deposited in the same repository on receipt, before in-principle acceptance. The 35-row structural sample described under Pilot data is deposited there for peer review, under embargo until Stage 2 acceptance, with a view-only link for the editors.

Code availability

All analysis code is version-controlled at https://github.com/pablobernabeu/InTRACt_Interplay_of_Text_Reader_And_Context, in the `preregistration` directory, which will be public, complete and available for peer review at submission. The directory contains the theory object with its checks and derived diagram, and the design specifications with the precision analysis and its recomputations at 0.95 assurance and under the Bayes-factor criterion. It also contains both versions of the count script supplied to the data controller and the scripts and record of the literature search. The theory object, the design analysis and the literature search can be reproduced from these files now. The outcome-model code described under Randomisation and blinding will be deposited in the same directory before the extract is transferred, together with the frozen feature inventory and the archived prior table.

The four packages the protocol relies on are cited by their version-independent identifiers. The releases in use are theoryforge 0.6.0 (Bernabeu, 2026d), pilotr 0.3.0 (Bernabeu, 2026b), scopusflow 0.4.0 (Bernabeu, 2026c) and depictr 0.3.0 (Bernabeu, 2026a). The software versions behind the reported numbers are recorded with their outputs, in `design_analysis/session_info.txt` and `design_analysis/session_info_ordinal.txt` for the design analysis, `theory/tf_check.txt` for the theory object and `literature/scopus_run_stamp.txt` for the literature search. The design-analysis figures were composed with depictr, which checks each panel's type size and greyscale legibility at the journal's print width and refuses to save a panel that fails.

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Acknowledgements

[Name all funding sources with grant numbers.] [Choose one: "The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript." / "The authors received no specific funding for this work."] We thank the data controller's team for building and holding the extract under the sequestration arrangement, and Nature Human Behaviour's editorial

office for advising on what we may receive before submission.

Author contributions

[Contributions in the CRediT taxonomy, by initials, to be added with the author list.]

Competing interests

[Choose one and complete. Either: “The data controller, Gameloft, provides the data under a data-sharing agreement with the University of Oxford and has no role in the design, analysis, interpretation or decision to publish; no author has received funding, employment or other consideration from Gameloft or Ferrero International. The authors declare no competing interests.” Or a statement describing any such relationship.]

Figures

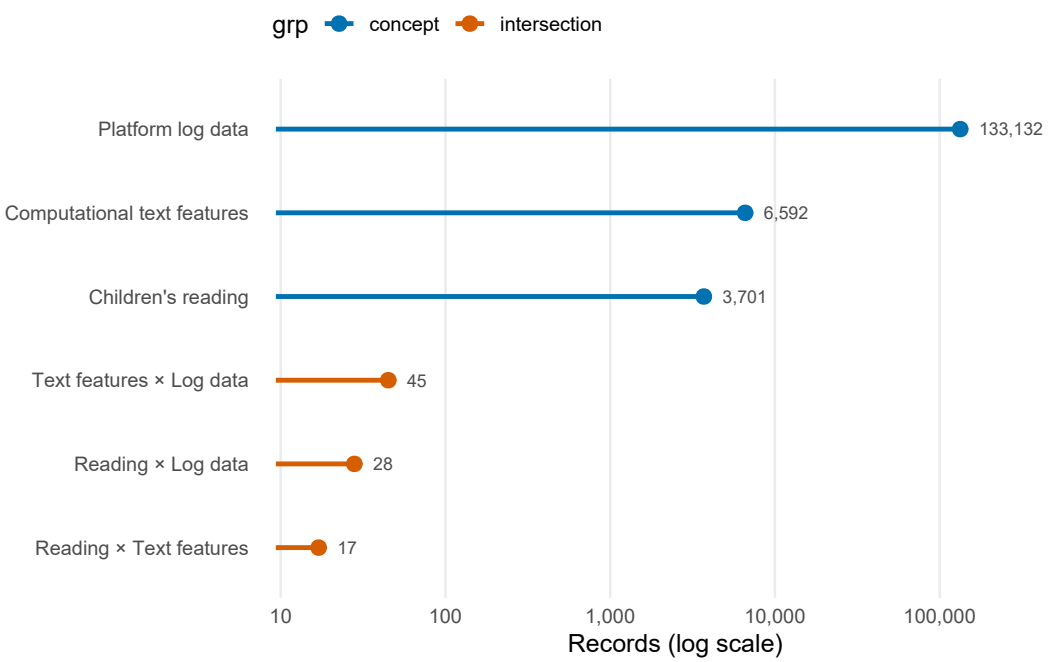


Figure 1

Size of the literature around this study. Scopus records from 2018 to the search date of 26 August 2026, for children's reading comprehension, computational text features and platform log data, and for every intersection of the three. The pairwise intersections hold 17, 28 and 45 records, and the three-way intersection holds none, so it does not appear on the logarithmic axis. Supplementary Section S4 gives the search expressions.

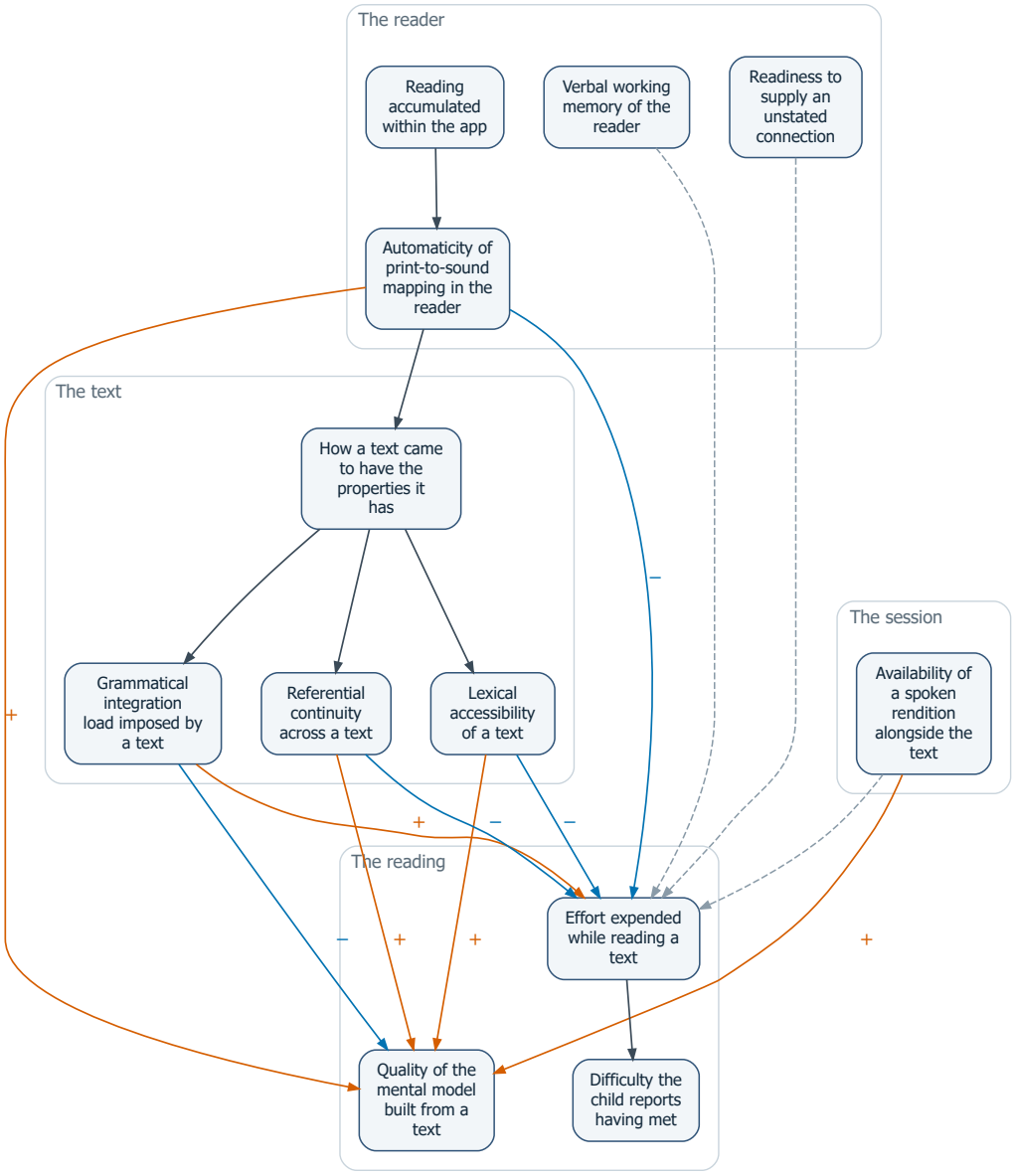


Figure 2

The nomological net, generated from the deposited theory object. Solid arrows are causal paths, labelled with the predicted sign where the theory fixes one, and dashed lines mark moderation. Text properties and reader capacities converge on the effort a reading requires, and that effort shapes the difficulty the child reports. Parallel paths lead to the quality of the child's mental model, which these data can measure only if comprehension items prove to exist. How a text came to have its properties (its generation route and target reading level) is a common cause of all three text properties and receives a path from the reader's decoding automaticity. Every text coefficient is therefore estimated within reading level.

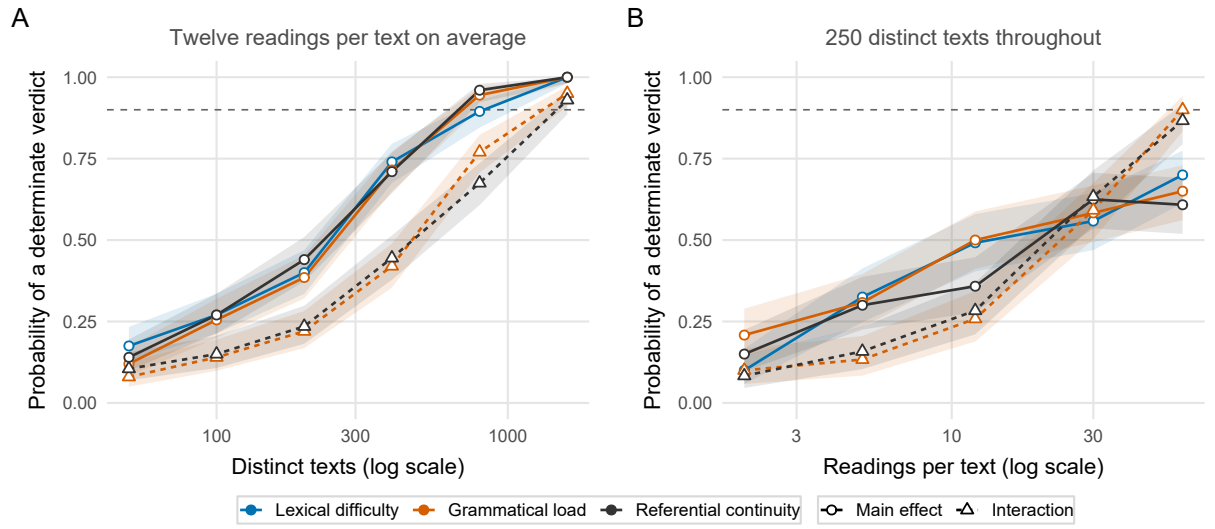


Figure 3

Precision by the number of distinct texts and by readings per text. Panel a varies the number of distinct texts at an average of twelve readings each. Panel b fixes the number of texts at 250 and varies readings per text, the quantity that accumulates most readily on this platform. Points show the probability that a 95 per cent interval falls wholly outside the region of practical equivalence when the simulated effect is 0.10 standard deviations. Probabilities are estimated from 200 simulation replicates per grid point in panel a and 120 in panel b. Shaded bands are 95 per cent Wilson score intervals on each probability, and the dashed line marks the 0.90 target.